



1.0 GROUND TEST PROCEDURES

WARNING

The pilot should inspect all appropriate maintenance records prior to any ground run or flight.

WARNING

A thorough preflight briefing and preflight inspection should be conducted prior to any ground run or flight.

1.1 EXTERNAL EMERGENCY LIGHTS

1. With no power (BATT – OFF) on the helicopter, place EXT EMERG light switch to ARM.
 - a. Verify all six white external emergency lights illuminate.

1.2 FUEL SELECTORS

WARNING

The FUEL SELECTOR switch must have a positive detent from XFD to OFF. The pilot must pull down on the switch to move the switch from XFD to OFF.

1. With no power (BATT – OFF) on the helicopter, place fuel selector switch for the #1 Engine in DIRECT and XFD.
 - a. Verify that the switch activates by listening for the valve to move.
 - b. Verify that each position has positive lock.
 - c. Verify that the switch can be moved from XFD to DIRECT with only forward pressure and vice versa.
 - d. Verify that the switch CANNOT be moved from XFD to OFF with ONLY aft pressure.
2. Repeat for #2 Engine.



1.3 STANDBY ATTITUDE INDICATOR

1. With no power (BATT – OFF) on the helicopter, push the EMER PWR pushbutton next to the standby attitude indicator.
 - a. Verify that the ARM and EMER PWR lights illuminate.
 - b. Verify that the beeping tone is heard in the cockpit.
 - c. Verify that the BATTERY TEST light illuminates.
2. Push the EMER PWR pushbutton.
 - a. Verify that the ARM and EMER PWR lights extinguish.
 - b. Verify that the beeping tone is silent.
 - c. Verify that the BATTERY TEST light extinguishes.

1.4 BATTERY CHECKS

1. Select BATT – ON.
 - a. Verify that only the pilot inboard and outboard MFDs are powered.
 - b. Verify internal communication among the pilot, copilot, external, forward ramp, and aft ramp positions are functional.
 - c. Verify that the copilot's RTU is powered.
 - d. Verify that the #1 communication radio is functional.
2. Rotate the TEST FIRE DET switch to the INBD position.
 - a. Verify that the master FIRE warning light illuminates on the pilot's master warning panel.
 - b. Verify that FIRE #1 ENG, FIRE #2 ENG and FIRE APU lights illuminate on the FIRE/ARM pushbuttons.
3. Rotate the TEST FIRE DET switch to the NORM position.
4. Push the EMER PWR pushbutton next to the standby attitude indicator.
 - a. Verify that the ARM and EMER PWR lights illuminate.
 - b. Verify that the beeping tone is heard in the cockpit.
 - c. Verify that the BATTERY TEST light illuminates.



- d. Verify that the standby airspeed, standby altimeter, standby attitude and standby compass lighting illuminates.
5. Push and hold the BATTERY TEST light for at least 5 seconds.
 - a. Verify that the GOOD light illuminates.
 - b. Verify that the BATTERY TEST light remains illuminated.
 - c. Verify that the EMER PWR and ARM lights are illuminated.
 - d. Verify that the beeping tone is silent.
 - e. Verify that the standby airspeed, standby altimeter, standby attitude and standby compass lighting extinguishes.
6. Release the BATTERY TEST light.
 - a. Verify that the GOOD light extinguishes.
 - b. Verify that the BATTERY TEST light remains illuminated.
 - c. Verify that the ARM and EMER PWR lights remain illuminated.
 - d. Verify that the beeping tone is heard.
 - e. Verify that the standby airspeed, standby altimeter, standby attitude and standby compass lighting illuminates.
7. Erect the gyro by pulling the PULL TO CAGE knob.
 - a. Verify that the gyro is erect with the attitude of the helicopter.
 - b. Verify that the red OFF flag disappears.
8. Cage the gyro by pulling PULL TO CAGE lever and then rotating it clockwise.
 - a. Verify that the red OFF flag is visible.
9. Push the EMER PWR pushbutton.
 - a. Verify that all the lights extinguish.
 - b. Verify that the beeping tone is silent.

1.5 SEATBELT AND SHOULDER HARNESS

1. Attach the pilot's seat belts and shoulder harness.
 - a. Verify that they are properly installed and easily adjusted.
2. Activate the inertia reel lock.



- a. Verify that the lock releases.
- b. Verify that the inertial lock engages with a quick pull of the shoulder harness.
3. Repeat for copilot's position.

1.6 SEAT ADJUSTMENT

1. Adjust the pilot seat through full range of travel both fore and aft, and up and down.
 - a. Verify that the seat moves smoothly.
 - b. Verify that the seats lock in position at any increment of travel when the levers are released.
2. Repeat for copilot's seat.

1.7 TAIL ROTOR PEDAL ADJUSTMENT

1. Pull the pilot pedals PED ADJ handle.
 - a. Verify that the pedals travel full range.
 - b. Verify that the lock holds the pedals at different positions throughout the full range.
2. With feet lightly on the pedals, verify that the pilot pedals travel to the full aft position when the PED ADJ is pulled due to spring action.
3. Repeat for copilot's pedals.

1.8 THROTTLE QUADRANT

1. Move the #1 ENGINE THROTTLE through full range.
 - a. Verify that the THROTTLE moves smoothly with some friction but does not bind.
 - b. Verify that friction holds the THROTTLE in position when released.
 - c. Verify that the IDLE and FLY detents engage positively.
 - d. Verify that the THROTTLE must be pushed in when moving aft out of the FLY position.
 - e. Verify that the THROTTLE must be pushed in when moving from IDLE to STOP.
2. Repeat for #2 ENGINE THROTTLE.



1.9 PLACARDS

1. Verify that the placard above the copilot's inboard MFD reads as follows:

THIS HELICOPTER MUST BE OPERATED IN ACCORDANCE WITH THE OPERATING LIMITS SPECIFIED IN THE FAA APPROVED ROTORCRAFT FLIGHT MANUAL. THE AIRWORTHINESS LIMITATIONS SECTION OF THE ROTORCRAFT MAINTENANCE MANUAL MUST BE COMPLIED WITH. . . .

2. Verify that a radio call placard is installed on the instrument panel.
3. Verify that an MGB Pump Failed placard is installed on the instrument panel.
4. Verify that the standby compass calibration card is installed and filled out.
5. Verify that the two AHRS compass calibration cards are installed and filled out.

1.10 EXTERNAL POWER

1. Verify that the helicopter can accept AC power.
2. Verify that the helicopter can accept DC power.

1.11 APU START

1. Verify that the MAINT COMP FAIL caution light is NOT illuminated.
2. Place the APU CONTR switch to the ON position.
 - a. Verify that the APU advisory light is illuminated when the APU is running normally.
3. Place the APU GEN switch to the ON position.
 - a. Verify that the following advisory lights are illuminated on the EICAS:

APU ON

APU GEN ON

APU ACCUM LOW (until APU accumulator is charged)

PRIME PUMP ON

HYD GRD PUMP ON

PARKING BRAKE (if set)



HYD 3 HI MODE

CHECK STATUS

- b. Verify that only the #3 hydraulic pressure indicator is in the green range (if the ground pump is ON).

1.12 PARKING BRAKE

1. Release the parking brake by depressing the pilot's brake pedals.
 - a. Verify that the PARKING BRAKE handle moves full down.
 - b. Verify that the PARKING BRAKE advisory light is extinguished.
2. Without depressing the brake pedals, pull up on the PARKING BRAKE.
 - a. Verify that the handle does NOT move.
 - b. Verify that the PARKING BRAKE advisory light is extinguished.
3. Reset the brakes by depressing the pedals and pulling the brake handle to full up.
 - a. Verify that the PARKING BRAKE advisory light is illuminated.
4. Repeat for copilot position.

1.13 ROTOR BRAKE

1. Place rotor brake ON.
 - a. Verify that the minimum pressure on the gage is 450 psi.
 - b. Verify that the ROTOR BRAKE caution light is illuminated.
2. Release the rotor brake.
 - a. Verify that the pressure is zero.
 - b. Verify that the ROTOR BRAKE caution light is extinguished.

1.14 HUMS

1. Press HLTH hard bezel.
2. Select HUMS.



3. Insert HUMS Data Transfer Memory Unit (DTMU) into the Data Transfer Unit mounted on the right hand avionics rack.
 - a. Verify that DTMU is accepted by noting on the MFD the percent of the card used.
4. Select ADMIN→DATE CHANGE→ENTR.
 - a. Verify the correct date.
5. If date is incorrect, use INCR, DECR, LEFT, RIGHT and ENTER to set the correct date.
6. Select ADMIN→TIME CHANGE→ENTR.
 - a. Verify the correct time.
7. If time is incorrect, use INCR, DECR, LEFT, RIGHT and ENTER to set the correct time.
8. Select ADMIN→ACFT CONFIG→ENTR.
 - a. Verify the aircraft model and serial number.
9. If aircraft model is incorrect, enter S-92A using INCR, DECR, LEFT, RIGHT and ENTER.
10. If aircraft serial number is incorrect, enter the correct serial number as 920XXX (i.e., 920057) using INCR, DECR, LEFT, RIGHT and ENTER.

1.15 CLOCK

1. Verify the clock indicates the correct local and UTC time.
2. Set clock if necessary.
3. In either GMT or LT mode, press and hold the SEL button.
 - a. Verify that the display test indicates "8888".
4. Select the FT (flight time) mode.
5. While in the FT mode, press and hold the CTL button.
 - a. Verify that the display indicates "9959".
6. Release the CTL button.
 - a. Verify that the display will reset to "0000".
7. Select the ET (elapsed time) mode.
8. Press the CTL button.



- a. Verify that the timer counts in seconds.
9. Press the CTL button.
 - a. Verify that the timer resets to "0000".

1.16 AIRCRAFT CONFIGURATION

1. Select the UTIL bezel key on the MFD.
2. Select the CONFIG bezel key.
3. Select the ACFT bezel key.
 - a. Verify that the CONFIG page represents the actual configuration of the helicopter.

1.17 FLIGHT MANAGEMENT SYSTEM

NOTE

ADC, DME and VOR navigation sensors are selected when the helicopter becomes airborne.

1. Select UNS-1 ON/OFF DIM button ON.
 - a. Verify that all FMS built in tests pass.
 - b. Confirm INIT 1/1 screen comes on after PASS indications for each subsystem.
 - c. Verify that the FMS navigation database is current.
2. Select any MSG and note discrepancies.
3. Select DATA 2/4.
 - a. Verify that the ADC, DME and VOR are DESEL (D).
4. Select GNSS1 or SBAS.
 - a. Confirm FMS receiving satellites.
5. Check the Bright/Dim functions.
6. Initialize the FMS to the present location.



1.18 MULTIFUNCTION FLIGHT DISPLAYS

1. Verify that the slip ball on each MFD is centered with the helicopter level.
2. Select the LUM control on each MFD.
 - a. Verify that the display can be adjusted from full bright to dark.
3. Select the PFD, NAV, EICAS, HLTH, and UTIL hard bezels on each MFD.
 - a. Verify appropriate displays are available on each MFD.
4. Select the PFD display on each MFD.
5. Select FULL, COMP, ARC, CAT A and HOVER (if available) soft bezels.
 - a. Verify appropriate displays are available on each MFD.
6. Select NAV hard bezel on each MFD.
7. Select FULL and ARC.
 - a. Verify appropriate displays are available on each MFD.

1.19 SLEW CONTROLLER

1. Select the NAV display on the pilot's MFD.
2. Move the thumb switch on the pilot's Slew Controller.
 - a. Verify that the cursor on the NAV display moves with the thumb switch.
 - b. Verify that the Latitude/Longitude of the cursor appears on the MFD.
3. Activate the index finger hook button on the Slew Controller to send the hook position to the FMS.
4. Select DTO→LIST on the FMS.
 - a. Verify that the hook position is stored as a waypoint (Rx) at the top of the LIST.
5. Repeat with the copilot's MFD and Slew Controller.

1.20 DISPLAY CONTROL PANEL

1. Rotate the pilot BRG 1 and 2 knobs from OFF, ADF, NAV and LNAV.
 - a. Verify that the bearing pointer on the pilot PFD changes and information is displayed on HSI.



2. Rotate REF SET knob through different settings:
 - a. HVR RA
 - b. BARO
 - c. ALTP
 - d. MIN
 - e. RA
 - f. TDP
 - g. VTDP
3. Pull up and rotate inner REF SET knob; ensure HVR RA, BARO, ALTP, MIN, RA, TDP and VTDP display changes on the PFD and CAT A display.
4. Rotate RANGE control knob.
 - a. Verify that the range on the PFD ARC, NAV FULL and NAV ARC displays change.
5. Select the CP/SC button.
 - a. Verify that the attitude display changes from cross pointer to single cue.
6. Select the NAV button.
 - a. Verify that the appropriate navigation source is displayed on bottom right side of PFD.
7. Select the LNAV button.
 - a. Verify that the appropriate LNAV navigation source is displayed on bottom right side of PFD.
8. Select the PREV button (3 times) while in LNAV mode.
 - a. Verify that the VOR 1, VOR 2 and P-ILS data appears within a dashed box above the LNAV navigation source label.
9. Select the LBS/KG button.
 - a. Verify that the fuel quantity toggles between LBS and KG.
10. Select the hPa/In Hg button.
 - a. Verify that the barometric altimeter setting toggles between hPa and in Hg.
11. Repeat for copilot Display Control Panel (DCP).



1.21 REMOTE INSTRUMENT CONTROLLER

1. Select NAV and VOR as source.
2. Rotate COURSE 1 knob.
 - a. Verify that course changes on PFD in HSI mode.
3. Center course 1 by pushing the COURSE 1 knob.
 - a. Verify that the #1 Course indicator centers if a navigation source is being received.
4. Repeat for COURSE 2.
5. Rotate HEADING knob.
 - a. Verify that the heading bug moves.
 - b. Verify that the new heading is digitally displayed on upper left side of HSI.
6. Press the top of the HEADING knob.
 - a. Verify that the heading synchronizes to present heading.
 - b. Verify that the digital heading disappears in approximately 5 seconds.

1.22 REVERSION CONTROL PANEL

NOTE

Yellow text indicates that the cross side source is driving the display.

NOTE

White text indicates that the on side source is driving the display.

NOTE

A Yellow Box around the text indicates that a reversion mode has been selected.

1. Place AHRS switch from NORM to 1.
 - a. Verify that AHRS 1 reversion annunciation is displayed in yellow on the pilot PFD (off side) and displayed in white on the copilot PFD (on side), each within a yellow box.
2. Place AHRS switch from NORM to 2.



- a. Verify that AHRS 2 reversion annunciation is displayed in yellow on the copilot PFD (off side) and displayed in white on the pilot PFD (on side), each within a yellow box.
3. Select AHRS NORM.
 - a. Verify that the AHRS reversion annunciations are removed from the PFDs.
4. Place the ADC switch from NORM to 1.
 - a. Verify that ADC 1 reversion annunciation is displayed in white on the copilot PFD (on side) and displayed in yellow on the pilot PFD (off side), each within a yellow box.
5. Place the ADC switch from NORM to 2.
 - a. Verify that ADC 2 reversion annunciation is displayed in yellow on the copilot PFD (off side) and displayed in white on the pilot PFD (on side), each within a yellow box.
6. Select ADC NORM.
 - a. Verify that the ADC reversion annunciations are removed from the PFDs.
7. Place DCU switch from NORM to 1.
 - a. Verify that DCU 1 reversion annunciation is displayed in yellow on the pilot PFD (off side) and displayed in white on the copilot PFD (on side), each within a yellow box.
8. Place DCU switch from NORM to 2.
 - a. Verify that DCU 2 reversion annunciation is displayed in white on the pilot PFD (on side) and displayed in yellow on the copilot PFD (off side), each within a yellow box.
9. Select DCU NORM.
 - a. Verify that DCU reversion annunciations are removed from the PFDs.

1.23 MFD REVERSION

1. Select PFD on pilot and copilot outboard MFDs.
2. Select non-PFD pages on pilot and copilot inboard MFDs.
3. Turn off the pilot MFD by pulling the following circuit breakers:
 - a. PLT MFD OBD – BATT HOLD UP Bus (overhead circuit breaker panel).
 - b. PLT OBD MFD – DC ESSENTIAL Bus (pilot circuit breaker panel).
 - c. Verify that the inboard pilot MFD automatically changes to the PFD.



4. Turn off the copilot MFD by pulling the following circuit breakers:
 - a. C/P OBD MFD – NO. 1 DC PRI Bus (copilot circuit breaker panel).
 - b. C/P OBD MFD – NO. 2 DC PRI Bus (pilot circuit breaker panel).
 - c. Verify that the inboard copilot MFD automatically changes to the PFD.

1.24 CAUTION/ADVISORY CHECK

NOTE

Caution light order may vary slightly depending on power up conditions and bus reporting sequencing.

1. Select EICAS page in the cockpit.
2. Select SHOW OBE soft bezel key.
 - a. Verify that the following warning and caution lights are illuminated, if applicable:

AC GEN 1 FAIL	MGB OIL PRES	AC GEN 2 FAIL
HYD 1 PUMP FAIL	MGB OIL OUT	HYD 2 PUMP FAIL
MGB PUMP 1 FAIL		MGB PUMP 2 FAIL
ENG 1 OIL PRES		ENG 2 OIL PRES
ENG 1 FUEL PRES		ENG 2 FUEL PRES
 - b. If the ground pump is not activated, verify that the following additional caution lights are illuminated:

	BOOST PRES	HYD 3 PUMP FAIL
SAS 1 PRES	MR 1 SVO PRES	SAS 2 PRES
	MR 2 SVO PRES	
	TR SVO 1 PRES	
	TR SVO 2 PRES	
3. Select HIDE OBE soft bezel key.
 - a. Verify that the above cautions are hidden.

1.25 AIRCRAFT GROSS WEIGHT

NOTE

A CHECK AGW advisory light will illuminate after takeoff if the pilots have not selected the AGW display. Selecting the display will extinguish the advisory light.



NOTE

Pilots' and passenger weights are entered as PASSENGER WEIGHT.

NOTE

AGW data is stored in both DCUs. DCU #1 is the master. If DCU #1 is replaced, AGW data will need to be re-entered. If DCU #2 is replaced, DCU #1 will automatically update DCU #2.

1. Press UTIL hard bezel.
2. Select AGW.
 - a. Verify the basic weight and make corrections as required.
 - b. Verify that fuel weight corresponds with fuel gages.
3. Input the correct pilot, passenger and cargo weights.

1.26 FIRE DETECTION TEST

NOTE

When engine cowlings are open, a valid test may not be possible.

NOTE

Once the VOICE CNCL button is pushed, the aural alert system will continue to speak until the current sentence is finished.

NOTE

Either "ENGINE #1" or "ENGINE #2" may be the first aural alert heard during the test.

1. Rotate the FIRE DET switch from NORM to INBD.
 - a. Verify that the pilot and copilot master FIRE warning lights illuminate.
 - b. Verify that the FIRE #1 ENG, FIRE #2 ENG and APU pushbutton lights illuminate.
 - c. Verify that "FIRE ENGINE ONE," "FIRE ENGINE TWO" and "FIRE APU" aural alerts are heard through the ICS.
2. Rotate the FIRE DET switch to NORM and wait for all the warning lights to extinguish.
3. Rotate the FIRE DET switch from NORM to OUTBD.
 - a. Verify that the pilot and copilot master FIRE warning lights illuminate.



- b. Verify that the FIRE #1 ENG and FIRE #2 ENG pushbutton lights illuminate.
 - c. Verify that "FIRE ENGINE ONE" and "FIRE ENGINE TWO" aural alerts are heard through the ICS.
4. Rotate the FIRE DET switch from OUTBD to SMOKE.
 - a. Verify that the SMOKE IN BAGGAGE warning illuminates on the EICAS page.
 - b. Verify that "SMOKE IN BAGGAGE" aural alert is heard through the ICS.
 5. Rotate the FIRE DET switch to NORM.

1.27 FIRE EXTINGUISHER TEST

NOTE

The APU and #1 engine are tested together and the #1 and #2 engines are tested together.

NOTE

The complete test can be conducted using AC external power without the APU or engines running.

NOTE

With only the APU running, the #1 engine and #2 engine can be tested.

NOTE

With the #2 engine running and the Main Generators online, the APU and #1 engine can be tested.

1. With the APU ON and the #1 and #2 engines not running, place the fuel control switches to DIR.
 - a. Verify that the green EICAS fuel lines are connected to the appropriate tanks and engines.
2. Place the fuel control switches to XFD.
 - a. Verify that the green EICAS fuel lines are connected to the appropriate tanks and engines.
3. Press the FIRE #1 ENG and FIRE #2 ENG guarded buttons on the fire control panel.
 - a. Verify that the ARM light illuminates on #1 engine and #2 engine buttons.
 - b. Verify that the EICAS fuel lines indicate OFF (disappear) to #1 engine and #2 engine.
4. Select the FIRE EXTG test switch to OPEN.



- a. Verify that the WARN light on the FIRE EXTG TEST panel (next to the switch) illuminates.
5. Activate the fire extinguisher select switch to the MAIN position.
 - a. Verify that the TEST light on the FIRE EXTG TEST panel illuminates.
6. Activate the fire extinguisher select switch to the RESERVE position.
 - a. Verify that the TEST light on the FIRE EXTG TEST panel illuminates.
7. Place the FIRE EXTG test switch to SHORT.
 - a. Verify that the WARN light and the TEST light on the FIRE EXTG TEST panel illuminate.
8. Activate the fire extinguisher select switch to MAIN position.
 - a. Verify that the WARN and TEST lights on the FIRE EXTG TEST panel remain illuminated.
9. Activate the fire extinguisher select switch to RESERVE position.
 - a. Verify that the WARN and TEST lights on the FIRE EXTG TEST panel remain illuminated.
10. Place the FIRE EXTG switch to NORM.
 - a. Verify that the WARN and TEST lights extinguish.
11. Depress both FIRE #1 engine and FIRE #2 engine guarded buttons on the fire control panel.
 - a. Verify that the ARM light extinguishes on #1 engine and #2 engine buttons.
 - b. Verify that green EICAS fuel lines appear on the EICAS.
12. With only the #2 engine running and the main generators on line, repeat the test for APU and the #1 engine.

1.28 ICS

NOTE

The EMER function on the ICS control allows the pilot and copilot a direct (hard) line to the #1 Radio. Internal ICS is inoperative when EMER is selected.

NOTE

The EMER function is only operational at the pilot and copilot positions.



NOTE

The pilot's audio controller controls what the forward ramp and aft ramp ICS stations monitors.

NOTE

The copilot's audio controller controls what the external ICS station monitors.

NOTE

The right cabin audio controller controls what the left cabin ICS station monitors (if installed).

1. Activate the pilot's audio controller.
 - a. Verify VOL control.
 - b. Verify cyclic switch operations.
 - c. Verify VOX operations.
 - d. Verify HOT operations.
 - e. Verify foot switch operations.
 - f. Verify EMER operations.
 - g. Verify that the forward and aft ramp ICS stations monitor the pilot's audio selections.
2. Repeat for the copilot's audio controller.
 - a. Verify that the external ICS station monitors the copilot's audio selections.
3. Repeat for the right cabin audio controller.
 - a. Verify that the left cabin ICS station monitors the right cabin audio selections.

1.29 LIGHTS

1.29.1 LANDING LIGHT

1. Place the LANDING NO 1 switch to ON.
 - a. Verify the left landing light illuminates.



2. Place the LANDING NO 2 switch to ON.
 - a. Verify the right landing light illuminates.
3. Place LANDING NO 1 and NO 2 switches to OFF.
 - a. Verify that both landing lights extinguish.

1.29.2 CONTROLLABLE SEARCHLIGHT

1. Place both the pilot's and copilot's SRCHLT switches on the collective to OFF.
2. Place the pilot's SRCHLT switch on the collective to ON.
 - a. Verify that the searchlight illuminates.
3. Extend, retract and swivel the searchlight using the SRCHLT SLEW controller on the collective.
 - a. Verify that the searchlight rotates through 360° and moves forward through a 120° arc from the stowed position.
4. Place the pilot's SRCHLT switch on the collective to STOW.
 - a. Verify that the searchlight stows in proper position.
5. Place the pilot's SRCHLT switch on the collective to OFF.
 - a. Verify that the searchlight extinguishes.
6. Repeat with the copilot's switches.

1.29.3 ANTI-COLLISION LIGHTS

1. Place the ANTI COLL UPPER switch to ON.
 - a. Verify that the upper Anti-Collision light illuminates.
2. Place the ANTI COLL LWR switch to ON.
 - a. Verify that the lower Anti-Collision light illuminates.
3. Place both ANTI COLL switches to OFF.
 - a. Verify that the Anti-Collision lights extinguish.



1.29.4 POSITION LIGHTS

1. Place the POSN switch to ON.
 - a. Verify that the red (left), green (right) and white (tail) position lights illuminate.
2. Place the POSN switch to OFF.
 - a. Verify that the three position lights extinguish.

1.29.5 COCKPIT LIGHTS CONTROL PANEL

1. Rotate all the cockpit light controls (BACK-UP IND, INSTR PNL, CONSOLE SIDE/OVHD/LWR, P/B DIMMER and FLOOD) on the overhead COCKPIT LIGHTS panel from OFF to BRT.
 - a. Verify that the lights illuminate.
 - b. Verify that the lights go from bright to dim.

1.29.6 PUSHBUTTON AND DIMMING LAMP TEST

1. Rotate the P/B DIMMER to OFF.
2. Hold the P/B LAMP switch to TEST.
3. Verify the following illuminate:
 - a. All pushbuttons on the AFCS control panel.
 - b. All 4 landing gear control panel lights and 2 filaments per light.
 - c. All lights on both master warning panels.
 - d. The 3 ARMED lights only on the engine fire control panel.
 - e. The IAS, ALT, HDG, and STBY lights on both mode select panels.
 - f. The ARMED, TEST and 2 L/R lights on the flotation control panel if installed.
 - g. The ARMED, TEST and L/R lights on the life raft panel if installed.
 - h. The ARMED, OFF, EXT PWR, GOOD and BATTERY TEST lights on the standby attitude indicator control.
4. Release the P/B LAMP switch.
 - a. Verify that the lights extinguish.



- b. Note the audio "DECOUPLE" on the ICS.
- 5. Rotate the P/B DIMMER from OFF to BRT.
 - a. Verify that the above lights dim and brighten.
- 6. Rotate to the OFF position.

1.29.7 BAGGAGE COMPARTMENT LIGHTS

- 1. Place the BAG COMP switch to ON.
 - a. Verify the baggage compartment light illuminates.
- 2. Place the BAG COMP switch to CREW.
- 3. On the auxiliary light control panel have the crew select BAG COMP ON.
 - a. Verify the baggage compartment light illuminates.
- 4. Place all baggage compartment lighting switches OFF.
 - a. Verify the baggage compartment light extinguishes.

1.29.8 CARGO LOADING LIGHT

- 1. Place the CARGO LOAD switch to ON.
 - a. Verify the cargo loading light illuminates.
- 2. Place the CARGO LOAD switch to CREW.
- 3. On the auxiliary light control panel have the crew select CARGO LOAD ON.
 - a. Verify that the cargo loading light illuminates.
- 4. Place all CARGO LOAD switches to OFF.
 - a. Verify that the cargo loading light extinguishes.

1.29.9 HEELS LIGHTING

- 1. Place the HEELS switch to ON.
 - a. Verify that the pilot and copilot cockpit HEELS lights illuminate.



2. Place the HEELS switch to OFF.
 - a. Verify that the pilot and copilot cockpit HEELS lights extinguish.

1.29.10 EXTERNAL EMERGENCY LIGHTS

1. Have the crew select OFF on the cabin control panel.
2. Place the EXT EMERG switch to ON.
 - a. Verify all six white external emergency lights illuminate.
3. Place the EXT EMERG switch to ARM.
 - a. Verify all six white external emergency lights extinguish.
4. On the cabin control panel, have crew select ON.
 - a. Verify all six white external emergency lights illuminate.
5. Place the cabin control panel switch to OFF.
6. Place the EXT EMERG switch to OFF.

1.29.11 UTILITY LIGHTS

1. On the pilot's utility light, rotate the intensity from OFF-DIM to BRIGHT.
 - a. Verify the light illuminates and goes from dim to bright.
2. Change the light pattern from FLOOD to SPOT.
3. Change the light color from RED to WHITE.
4. Press the Morse Code button.
 - a. Verify the light illuminates at full intensity when the button is pressed.
5. Rotate the intensity to OFF.
6. Repeat for copilot's utility light.



1.30 PITOT-STATIC AND OAT

1.30.1 BAROMETRIC ALTIMETER

1. Obtain the current barometric altimeter setting from an outside source.
2. Adjust the standby altimeter to the current altimeter setting.
3. Input the current setting into the pilot and copilot MFD via the DCP by selecting BARO and adjusting the set knob.
 - a. Verify that all three helicopter's altimeters are within 70 feet of field elevation.
 - b. Verify that the pilot, copilot and standby altimeters are within 50 feet of one another.

1.30.2 VERTICAL VELOCITY INDICATOR

1. Verify that the pilot and copilot VVI indicates zero vertical velocity when the helicopter is stationary on the ground.

1.30.3 AIRSPEED INDICATORS

NOTE

The airspeed indicators will indicate approximately 28 KIAS on the ground with no wind.

1. Verify that the pilot and copilot PFDs display a valid airspeed.
2. Verify that the backup airspeed indicator displays a valid airspeed.

1.30.4 OAT GAGE

NOTE

The OAT sensors for the ADC are located under the helicopter. The OAT sensors for the FADEC are located near the engine inlets. The temperature between the top and bottom of the helicopter may vary slightly.

1. Obtain an OAT from an outside source.
 - a. Verify that the OAT displayed on the pilot and copilot PFDs are $\pm 1^{\circ}\text{C}$ from the outside source.



- b. Verify that the OAT displayed on the EPAC page is $\pm 1^{\circ}\text{C}$ from the outside source.

1.31 VHF RADIO

1. From the pilot position, functionally check the #1 VHF radios with the tower.
 - a. Verify the cyclic switch is operational.
 - b. Verify the foot switch is operational.
2. Repeat for #2 VHF radio.
3. Repeat for copilot position.

1.32 GUARD RADIO

NOTE

Switch must be in NORMAL to transmit.

1. Select NO. 1 RADIO GUARD switch to GUARD, then NORMAL.
 - a. Verify that the #1 radio switches to 121.5.
2. Retune the #1 radio to the desired frequency.

1.33 ENHANCED GROUND PROXIMITY WARNING SYSTEM

NOTE

The FMS must be ON with a good current position to complete a valid EGPWS IBIT.

NOTE

If ILS NAV is selected as the navigation source and the selected course is set for back course (B/C), the "GLIDESLOPE" aural alert will not activate during EGPWS IBIT.

1. Select PFD ARC on the pilot and copilot outboard MFDs.
2. Select NAV on the pilot's inboard MFD.
3. Overlay the terrain on both the NAV and PFDs.



4. Select the HLTH page on the copilot inboard MFD.
5. Select AVIONICS, SUB STAT and EGPWS.
6. Press the IBIT bezel key on the left side of the MFD.
 - a. Verify that the amber EGPWS INOP and white TER TEST illuminate on the PFD and remain on for the duration of the test.
 - b. Verify that the white LOW ALT illuminates momentarily on the NAV page and PFD.
 - c. Verify that the "GLIDESLOPE" aural warning is annunciated in the headset.
 - d. Verify that the amber G/S CANCELLED illuminates momentarily on the NAV page and PFD ARC.
 - e. Verify that the "PULL UP" aural warning is annunciated in the headset.
 - f. Verify that the "WARNING TERRAIN" aural warning is annunciated in the headset.
 - g. Verify that the terrain display self-test pattern is displayed on the NAV page and the PFD ARC. The terrain display test pattern will be removed after several sweeps.
 - h. Verify that the GPWS-INOP and TER TEST labels are removed from the display at the completion of the test.
 - i. Verify that the TEST PASSED status is displayed on the HLTH page.

1.34 COCKPIT VOICE RECORDER/FLIGHT DATA RECORDER

NOTE

The CVR/FDR has an independent power supply that continues recording 10 minutes after loss of aircraft power. Therefore, the CVR/FDR will take 10 minutes to change status to INOPERATIONAL after initiating the test.

1. Select the HLTH page.
2. Select AVIONICS→SUB STAT→CVR/FDR.
 - a. Verify that the CVR/FDR status is OPERATIONAL.
3. Select IBIT.
 - a. Verify that the CVR/FDR status changes to TEST IN PROGRESS.
 - b. Verify that the CVR/FDR status changes to TEST PASS.
 - c. Verify that the CVR/FDR status changes to OPERATIONAL.



4. Select and momentarily hold IMRS on the CVR/FDR switch.
 - a. Verify that the CVR/FDR status changes to INOPERATIVE after 10 ± 1 minutes.
5. Select OFF on the CVR/FDR switch.
6. Push the CVR/FDR RESET button for at least 2 seconds.
 - a. Verify that the CVR/FDR status changes to OPERATIONAL.
7. Select and momentarily hold INERTIA on the CVR/FDR switch.
 - a. Verify that the CVR/FDR status changes to INOPERATIVE after 10 ± 1 minutes.
8. Select OFF on the CVR/FDR switch.
9. Push the CVR/FDR RESET button for at least 2 seconds.
 - a. Verify that the CVR/FDR status changes to OPERATIONAL.

1.35 RADAR ALTIMETER

NOTE

An altitude of more than ± 5 feet on the ground will display OUT OF CAL as a CURR FAULT. The radar altimeter takes approximately 10 minutes to warm up and give the proper reading on the ground.

1. Select the HLTH page.
2. Select AVIONICS→SUB STAT→PAGE DOWN→RADALT 1.
 - a. Verify that the RADALT status is OPERATIONAL.
3. Select IBIT.
 - a. Verify that the RADALT status changes to TEST IN PROGRESS.
 - b. Verify that the radar altimeter display momentarily goes to 100 ft ± 10 and two blue Ts display during the test.
 - c. Verify that the RADALT status changes to TEST PASSED.
 - d. Verify that the RADALT status changes to OPERATIONAL.
4. Repeat for RADALT 2, if installed.



1.36 HYDRAULIC LEAK ISOLATION TEST

1. Place the rotor brake ON.
2. Place the GND PUMP switch to ON.
3. Select the HLTH page.
4. Select HYDRAULICS.
5. Select LDI TEST.
6. On the HLTH page:
 - a. Verify that the HYDRAULIC status changes to TEST IN PROCESS.
 - b. MFD will display various HYD failure indications.
 - c. Verify that the LDI indicated TEST PASSED.
 - d. Verify the HYDRAULIC status changes to OPERATIONAL.

1.37 FLIGHT CONTROL LIMIT CHECK

NOTE

This is a pilot's ground flight control limit check. It is designed to verify that the controls hit the proper control stops or limiters. It does not replace the ground maintenance rigging check in the maintenance manuals. Each aircraft may be slightly different, but the movement of the flight controls as indicated in these procedures is a good indication that the control stops and limiters are set properly, that there are no unwanted restrictions to the flight controls and that the pilot has full range. However, these procedures do not completely verify that the aircraft is properly rigged.

NOTE

The flight control system contains primary and secondary stops as well as limiters to restrict the range of motion of the swashplate and tail rotor servo. The cyclic stick stops are bolt stops located at the base of the pilot and copilot cyclic. The collective stick stops are bolt stops located at the base of the pilot and copilot collective. The yaw pedal stops are bolt stops located forward of the pilot and copilot pedals. The remaining stops are located and defined below:

Channel	Primary Stop	Secondary Stop
Collective	High – Collective sticks	High – Mixer
	Low – Mixer	Low – Collective sticks



Pitch	Cyclic Stick	SAS/Boost output arm
Roll	Cyclic Stick	SAS/Boost output arm
Yaw	Fwd quadrant	Yaw pedals

NOTE

The two collective to pitch limiters are located in the mixer. These limiters restrict the pitch inputs at the rotor for the high and low collective positions.

1.37.1 PEDALS

1. Select GND PUMP ON.
2. Center cyclic.
3. Lower the collective full down.
4. Apply full left pedal.
 - a. The left pedal should contact the left bolt stop in the cockpit.
5. Center pedals.
6. Raise the collective full up.
7. Apply full right pedal.
 - a. The right pedal should contact the right bolt stop in the cockpit.
8. Center the pedals.
 - a. Verify that the pilots and copilots pedals are centered and matched.

1.37.2 COLLECTIVE

1. Center cyclic and pedals.
2. Lower the collective full down.
 - a. The collective should contact the collective mixer stop. However, with a little down pressure the system may deflect enough for the collective to contact the lower stick stop.
3. Raise the collective full up.
 - a. The collective should contact the collective upper stick stop.



1.37.3 CYCLIC LONGITUDINAL

1. Center cyclic and pedals.
2. Lower the collective to the full down position.
3. Move the cyclic to the full forward position and maintain slight forward pressure on the cyclic.
4. Slowly raise collective to full up.
 - a. Cyclic must move forward as the collective goes from bottom to mid position indicating operation of the low forward limiter.
 - b. At approximately mid collective, the cyclic must contact the forward cyclic stop.
 - c. As the collective is moved up from mid to the high stop, the cyclic may move aft indicating operation of the high forward limiter.
5. If the stick does not move aft slightly to indicate operation of the high forward limiter, slowly apply full right pedal.
 - a. The cyclic should move aft indicating operation of the high forward limiter.
6. With the collective full up and the pedals centered, move the cyclic to the full aft position.
7. Maintain slight aft pressure on the cyclic.
8. The cyclic may not reach the aft cyclic stop.
9. Slowly apply full left pedal.
 - a. Cyclic should move slightly forward, indicating operation of the high aft limiter.
10. Center the pedals.
11. Slowly lower collective.
 - a. Initially, the cyclic may move slight aft and contact the aft cyclic stop (if it had not reached the aft stick stop at high collective).
 - b. At approximately collective mid position, the cyclic must start moving forward indicating operation of the low aft limiter.



1.37.4 CYCLIC LATERAL

1. With the pedals centered and collective full down, move the cyclic full left.
 - a. The cyclic should contact the left cyclic stop.
2. Move the cyclic full right.
 - a. The cyclic should contact the right cyclic stop.

1.38 PRIMARY SERVOS CHECK

1. Select GND PUMP ON.
2. Select BST ON.
3. Select SAS 1 and SAS 2 OFF.
4. Select YAW/CYCLIC/COLL TRIM OFF.
5. Place the SERVO switch to the BOTH position.
6. Cyclic and pedals centered.
7. Place the servo switch to NO. 1 position.
 - a. Verify that the MR SVO 2 PRES and TR SVO 2 caution lights illuminate.
 - b. Verify that the cyclic stick does not jump when securing the primary servo.
8. Slowly move the flight controls through full range.
 - a. Check for unusual binding or restrictions.
9. Place the servo switch to NO. 2 position.
 - a. Verify that the MR SVO 1 PRES and TR SVO 1 caution lights illuminate.
 - b. Verify that the cyclic stick does not jump when securing the primary servo.
10. Slowly move the flight controls through full range.
 - a. Check for unusual binding or restrictions.
11. Place the SERVO switch to the BOTH position.
 - a. Confirm all servo caution lights extinguish.
12. Slowly move the flight controls through full range.



- a. Check for unusual binding or restrictions.

1.39 BOOST SERVOS CHECK

NOTE

With boost OFF there is an interaction between the collective and the pedals.

1. Select GND PUMP ON.
2. Select BST OFF.
 - a. Verify that the BOOST PRES caution light illuminates.
3. Slowly move flight controls through full range.
 - a. Verify that there is no unusual binding, restriction or excessive dead band.
 - b. Note increased forces.
4. Select BST ON.

1.40 TRIM CHECKS

1. Select GND PUMP ON.
2. Select BST ON.
3. Select YAW, CYCLIC and COLL TRIM ON.
4. Select ATT, AP 1, AP 2, SAS 1 PRI and SAS 2 PRI ON.

1.40.1 DISENGAGEMENT CHECK

1. Center controls.
2. Select CYCLIC TRIM OFF.
 - a. Verify that the cyclic does not move.
3. Select CYCLIC TRIM ON.
4. Repeat for collective and pedals.



1.40.2 TRIM RELEASE CHECK

1. Depress cyclic TRIM release switch and move cyclic slowly.
 - a. Verify that the trim disengages.
2. Repeat for collective and pedals.

1.40.3 FORCE GRADIENT CHECK

1. Without pressing cyclic TRIM release, move the cyclic fore/aft and left/right.
 - a. Verify that a force gradient exists, i.e., increasing force with displacement.
2. Repeat for collective and pedals.

1.40.4 BEEPER TRIM CHECK

1. On the pilot's cyclic, activate the cyclic beeper trim in the fore/aft and lateral directions.
 - a. Verify that the beeper trim operates smoothly.
2. Repeat for collective and pedals.

1.40.5 TRIM RETURN CHECK

1. Center cyclic.
2. Without depressing the trim release, displace cyclic full fore/aft and left/right.
 - a. Verify that the cyclic returns smoothly to the center position.
3. Without depressing the pedal micro-switches, displace either pedal to full forward and release.
 - a. Verify that the pedals return smoothly to the neutral position.
4. Repeat with the other pedal.
5. From collective mid position, without pressing the collective trim switch, slowly displace the collective full up and release.
 - a. Verify that the collective returns smoothly to mid-position.



6. Repeat for collective full down.

1.40.6 COPILOT'S CONTROLS

1. Check copilot's cyclic, collective and pedal TRIM release switch for proper operation.
2. Check copilot's cyclic and collective beeper trim for proper operation.

1.41 AFCS PREFLIGHT BIT CHECK

WARNING

Prior to initiating PFLT BIT, ensure that all personnel are clear of all flight control systems.

NOTE

Return all flight controls to their original prestart condition prior to securing the ground check pump. Collective should be about mid position, with cyclic and pedals centered. Failure to do so could result in the transfer of hydraulic fluid from systems 1 and 2 into system 3.

1. Center all flight controls using the HUMS flight control position sensors. (HLTH→HUMS→MAINT→PARAMS→ENTR→ANLG→ANLG 2).
2. SAS 1 PRI and SAS 2 PRI – ON.
3. AP 1 and AP 2 – ON.
4. GND PUMP – ON.
 - a. Assure HYDRAULIC status is OPERATIONAL.
5. BST – ON.
6. Rotor brake – ON.
7. CYCLIC, YAW and COLL TRIM – ON.
8. Ensure ATT mode – ON.
9. AP 1 and AP 2 AP PWR RESET – Press and hold for two seconds.
10. Press hard bezel TEST on either pilot or copilot FD MODE SEL panel.



11. Press and hold PFLT BIT soft bezel for at least 0.5 seconds but no more than 2 seconds.
 - a. Follow instructions on the FD MODE SEL panel while PFLT is running.
 - b. Monitor progress on the HLTH page.
 - c. If any fault codes are displayed, record and notify maintenance.
 - d. If no fault codes are displayed – TEST complete.
12. Press and hold the AP 1/2 PWR RESET buttons for 2 seconds.
13. Trim controls to their center position.
14. GND PUMP – OFF.

1.42 WINDSHIELD WIPER/WASHER



Do not activate the windshield wipers on a dry windshield.

NOTE

Standoff pins shall be inserted prior to this check.

1. Place the WDSHLD WASH switch ON.
 - a. Verify that fluid flows from all the holes in the pilot and copilot windshield spray bars.
2. Place the WDSHLD WASH switch OFF.
 - a. Verify that fluid stops flowing.
3. Place the WDSHLD WIPER switch in SLOW.
 - a. Verify that the wipers operate at slow speed.
4. Place the WDSHLD WIPER switch in FAST.
 - a. Verify that the wipers operate at fast speed.
5. Place the WDSHLD WIPER switch OFF.
 - a. Verify that the wipers park parallel to the windshield inboard edge and about 1 inch from the edge.



1.43 PITOT HEATER

WARNING

The pitot tubes can become very hot during the test.

NOTE

The TURN OFF PITOT HT caution light will illuminate if the OAT is above 41°F and any pitot heat is selected ON.

1. Select the PITOT HT CPLT, PLT and STBY switches ON.
2. Pull the three PITOT HT PWR circuit breakers.
 - a. Verify that the PITOT 1 HEAT, PITOT 2 HEAT and PITOT 3 HEAT caution lights illuminate.
3. Reset the three circuit breakers.
 - a. Verify that the caution lights extinguish.
 - b. Have crewman verify that pitot tubes are heated.
4. Select PITOT HT CPLT, PLT and STBY switches OFF.

1.44 ENVIRONMENTAL CONTROL SYSTEM

NOTE

The ECS may be checked with the APU only or both engines running.

1. Place the AIR SRCE HEAT/STRT switch in APU.
2. Select CABIN VENT.
 - a. Verify that vent air flows from upper registers.
3. Select CABIN HEAT and rotate the cabin rheostat full clockwise to H.
 - a. Verify that warm air flows from all lower registers in the cabin.
4. Select CABIN HT and VT.
 - a. Verify that warm air flows from all lower registers and vent air flows from upper registers.



5. Select CABIN AC and rotate the cabin rheostat full counter clockwise to C.
 - a. Verify that cold air flows from upper registers.
6. Select CABIN MAX AC.
 - a. Verify that cold air flows from upper registers.
7. Repeat for cockpit.



2.0 RUN-UP TEST PROCEDURE

2.1 STARTER SYSTEM

2.1.1 START SWITCH

NOTE

The following steps confirm that pressing the START switch after the starter is engaged will disengage the starter.

NOTE

If the OAT is hotter than requested by the heater rheostats, no air will flow from the heater registers.

1. Place AIR SRCE HEAT/START switch in APU.
2. Place the CABIN and COCKPIT heater switches in HEAT.
3. Rotate the heater rheostats full clockwise to H.
 - a. Verify that air flows from the heater vents, if hot air is required.
4. Select NO. 1 FUEL selector OFF.
5. Select ENG IGN OFF.
6. Select NO. 1 THROTTLE OFF.
7. Press the NO. 1 ENG START button.
 - a. Verify that the STARTER 1 ON advisory light illuminates.
 - b. Verify that no air flows from the heater registers.
8. Continue motoring the engine until Ng stabilizes.
 - a. Verify Ng (24% Ng minimum).
9. Press the NO. 1 ENG START button to abort the start.
 - a. Verify that the STARTER 1 ON advisory light extinguishes.
 - b. Verify that #1 engine Ng decreases.
 - c. Verify that air flows from the heater vents, if hot air is required.



10. Repeat for the #2 engine.
11. Place the HEAT switch OFF.

2.1.2 THROTTLE DISENGAGEMENT

NOTE

The following steps confirm that moving the THROTTLES from IDLE to OFF will disengage the starter.

1. Select NO. 1 FUEL selector OFF.
2. Select ENG IGN OFF.
3. Place the NO. 1 THROTTLE in IDLE.
4. Press the NO. 1 ENG START switch.
 - a. Verify that the STARTER 1 ON advisory light illuminates.
 - b. Verify that #1 engine Ng increases.
5. Move NO. 1 THROTTLE from IDLE to OFF.
 - a. Verify that the STARTER 1 ON advisory light extinguishes.
 - b. Verify that the #1 engine Ng decreases.
6. Repeat for the #2 engine.

2.2 ENGINE START



Engine start will be aborted for any of the following reasons:
TGT exceeds 1000°C, no indication of oil pressure within 30 seconds, light off does not occur within 20 seconds after initial indication of Ng or throttle advancement to IDLE, or starter drops out prior to 50% Ng.

**CAUTION**

With an engine inlet removed, the helicopter is restricted to flat pitch operations. For dual engine operations with either or both inlets removed, the helicopter is restricted to one engine in IDLE and the other engine in FLY. Defer all engine checks (accel/decel) until the High Speed Shaft (HSS) is balanced and the engine inlets are installed.

NOTE

This procedure should be used when a different engine is installed or an engine has been removed and reinstalled.

NOTE

To collect HSS data to balance a shaft, the inlet on the engine being balanced must be removed to attach the optical sensor.

1. Select HIDE OBE on the EICAS page if displayed. The intent is to hide unnecessary cautions.
2. Place ROTOR BRAKE ON.
 - a. Verify at least 450 psi.
 - b. Verify that the ROTOR BRAKE caution light illuminates.
3. Place the ENG IGN switch in AUTO.
4. Place the fuel selector switch for the #1 engine to DIR.
 - a. Verify that the green EICAS fuel lines changes from OFF to DIR.
5. Press the NO. 1 ENG START switch.
 - a. Verify that the STARTER 1 ON advisory light illuminates.
 - b. Confirm TGT below 150°C.
6. Advance throttle to IDLE:
 - a. Verify that the ENG 1 FUEL PRES caution light extinguishes.
 - b. Verify that the engine oil pressure is in the green.
 - c. Verify that the ENG 1 OIL PRES caution light extinguishes.
 - d. Verify that the STARTER 1 ON advisory light extinguishes.



- e. Verify that Ng stabilizes at IDLE.
7. Run the engine 1 minute in DIR.
8. Place the fuel selector for the #1 engine to XFD.
 - a. Verify that the green EICAS fuel lines changes from DIR to XFD.
9. Run the engine 1 minute in XFD.
10. Retard throttle to OFF.
11. Repeat for #2 engine.

2.3 ENGINE CROSSBLEED START

CAUTION

The pilot must maintain 90% or higher Ng on the operating engine during a crossbleed start. Less than 90% Ng may not provide sufficient air for a successful crossbleed start and may result in a hung or hot start.

NOTE

Regardless of AIR SRCE HT/START switch position, if the APU is operating during an attempted crossbleed start, the APU may be providing the air source and not the opposing engine. The APU must be OFF to successfully complete a crossbleed start.

NOTE

This procedure should be used when a different engine is installed or an engine has been removed and reinstalled.

NOTE

Even if only one engine has been installed, a crossbleed start from both engines should be conducted.

1. Start one engine and set throttle to FLY.
2. Set Nr at 105%.
3. Ensure #1 and #2 GEN ON.



4. Place the APU CONTR switch to the OFF position.
5. Place the AIR SRCE HEAT/START switch to ENG.
6. Throttle – OFF (engine to be started).
7. FUEL SELECTOR – DIR.
8. Adjust collective to maintain a minimum of 90% Ng on the operating engine.
9. Press the ENG STARTER switch for the engine being started.
10. Continue motoring the engine until Ng peaks.
 - a. Verify Ng at least 24%.
11. Place the throttle to IDLE when TGT $\leq 150^{\circ}\text{C}$ and Ng 24%.
12. Monitor normal start.
13. Repeat for other engine.

2.4 ENGINE CROSSBLEED MOTORING



The pilot must maintain 90% or higher Ng on the operating engine during a crossbleed motoring. Less than 90% Ng may not provide sufficient air for Ng to accelerate to 24%.

NOTE

Regardless of AIR SRCE HT/START switch position, if the APU is operating during an attempted crossbleed start, the APU may be providing the air source and not the opposing engine. The APU must be OFF to successfully complete a crossbleed motoring check.

NOTE

This procedure should be used when a different engine is installed or an engine has been removed and reinstalled.

NOTE

Even if only one engine has been installed, a crossbleed motoring check from both engines should be conducted.



1. Start one engine and set throttle to FLY.
2. Set Nr at 105%.
3. Ensure #1 and #2 GEN ON.
4. Place the APU CONTR switch to the OFF position.
5. Place the AIR SRCE HEAT/START switch to ENG.
6. Throttle – OFF (engine to be motored).
7. FUEL SELECTOR – OFF (engine to be motored).
8. Adjust collective to maintain a minimum of 90% Ng on the operating engine.
9. Press the ENG STARTER switch for the engine being motored.
10. Continue motoring the engine until Ng peaks.
 - a. Verify that Ng is at least 24%.
11. Disengage starter.
12. Repeat for other engine.

2.5 ROTOR ENGAGEMENT



If engines starts were made with the rotor brake ON, remain at 50% to 54% Nr for a minimum of 1 minute prior to advancing the throttles to FLY.

1. Adjust Nr to 50 – 55% with the throttles.
 - a. Verify that MGB OIL PRES caution light extinguishes.
 - b. Verify that the main gear box oil pressure is in the green (minimum 30 psi).
 - c. Verify that the HYD 1/2/3 PUMP FAIL caution lights extinguish.
 - d. Verify that the hydraulic pressure in all three systems are in the green.
 - e. Verify that all servo pressure caution lights are extinguished.



2.6 HYDRAULIC SYSTEM CHECK

1. Place the HYDRAULICS SERVO switch to NO. 1.
 - a. Verify that the MR SVO 2 PRES caution light illuminates.
 - b. Verify that the TR SVO 2 PRES caution light illuminates.
2. Place the HYDRAULICS SERVO switch to BOTH.
 - a. Verify caution lights extinguish.
3. Place the HYDRAULICS SERVO switch to NO. 2.
 - a. Verify that the MR SVO 1 PRES caution light illuminates.
 - b. Verify that the TR SVO 1 PRES caution light illuminates.
4. Place the HYDRAULICS SERVO switch to BOTH.
 - a. Verify caution lights extinguish.

2.7 INITIAL RUN-UP CHECKS

1. Advance both throttles to FLY.
 - a. Verify that the droop stops are out by 70% Nr.
 - b. Verify that Nr is set to 105% Nr.
 - c. Verify that the MGB oil pressure is between 55 and 60 psi.
 - d. Verify that the hydraulic and engine instruments on the EICAS page are in the green.
 - e. Verify that the aircraft exhibits no unusual vibration.
 - f. Verify that #1 and #2 torques are matched within 5%.

2.8 MGB OIL BYPASS CHECK

NOTE

A rise in MGB oil pressure and temperature may accompany the MGB Oil Bypass Check.



NOTE

The auto bypass IBIT will be conducted as Nr passes through 60%.

1. At 105% Nr, place the MGB OIL BYP switch from AUTO to BYP.
 - a. Verify that the MGB BYPASS caution light illuminates.
2. Place the MGB OIL BYP switch in AUTO.
 - a. Verify that the MGB BYPASS caution light extinguishes.
3. Place the MGB OIL BYP switch in MAN COOL.
 - a. Verify that the MGB MAN COOL caution light illuminates.
4. Place the MGB OIL BYP switch in AUTO.
 - a. Verify that the MGB MAN COOL caution light extinguishes.

2.9 DRIVE SYSTEM LEAK CHECK

NOTE

For any MGB input module maintenance leak check or an engine input/output shaft seal replacement, start and run-up using the appropriate side engine.

NOTE

In extremely cold weather, MGB oil temperature above 70°C might not be possible.

1. Start engine.
2. Advance throttle to FLY.
3. Ground run for a minimum of 5 minutes at 105% Nr. If possible, ensure that the MGB oil temperature is in excess of 70°C and the MGB oil pressure is in the normal operating range.
4. Shut down helicopter.
5. Have maintenance conduct visual leak check.



2.10 ENGINE ACCELERATION CHECK

CAUTION

The engine inlets must be installed prior to conducting the Engine Acceleration Check.

1. Both THROTTLES – FLY.
2. Set Nr/Np – 105%.
3. THROTTLE 1 – Retard to IDLE and allow engine parameters to stabilize.
4. THROTTLE 1 – Advance rapidly to FLY (in one second) and hold until Ng peaks.
5. THROTTLE 1 – Retard rapidly to IDLE (no stalls should occur).
6. THROTTLE 1 – Return to FLY position.
7. Repeat for the other engine.

2.11 ENGINE BREAK-IN

NOTE

The following procedure should be used after a gas generator rotor and stator assembly has been changed in the field. When done properly, the break-in ensures that the gas generator section will operate reliably and with the highest efficiency. If the break-in procedure is neglected or is not accomplished properly, the gas generator efficiency may deteriorate, resulting in performance loss and an increase in specific fuel consumption.

NOTE

This procedure is more easily accomplished at higher aircraft gross weights. Consider using full fuel tanks and approved internal loads.

NOTE

High speed shaft balancing must be accomplished before the engine break-in procedure.

NOTE

The engine inlets must be installed during the break-in procedure.

**CAUTION**

Do not tether the aircraft to the ground to accomplish this procedure.

1. Start non break-in engine.
2. With collective full down, advance throttle of non-break-in engine to FLY.
3. Confirm $N_r = 105\%$.
4. Start break-in engine.
5. With collective full down, slowly advance throttle of break-in engine to FLY.
6. Slowly retard throttle of non-break-in engine until the N_g of the break-in engine reaches 85% or the throttle is in IDLE.
7. Plunge down on the collective OEI switch to select continuous OEI power.
8. If required, slowly increase collective until N_g of break-in engine is 85% ($\pm 1\%$).
9. Remain at 85% N_g on break-in engine for 30 seconds.
10. Ensure the throttle of the non-break-in engine is in IDLE.
11. Slowly increase collective until N_g of break-in engine is 90% ($\pm 1\%$).
12. Remain at 90% N_g for 30 seconds.
13. Slowly reduce collective until N_g of break-in engine is 85% ($\pm 1\%$).
14. Remain at 85% for 30 seconds.
15. Slowly increase collective until N_g of break-in engine is 90% ($\pm 1\%$).
16. Remain at 90% N_g for 30 seconds.

NOTE

Continue to 95% N_g only if this can be accomplished without lifting to a hover.

17. Slowly increase collective until N_g of break-in engine is 95% ($\pm 1\%$).
18. Remain at 95% N_g for 30 seconds.
19. Slowly reduce collective until N_g of break-in engine is 90% ($\pm 1\%$).



20. Remain at 90% Ng for 30 seconds.
21. Slowly increase collective until Ng of break-in engine is 95% ($\pm 1\%$). █
22. Remain at 95% for 30 seconds.

NOTE

The minimum attained Ng to complete this procedure is 95%. The pilot should attempt to reach Ng's above 95% to improve future engine performance.

23. Slowly increase collective to maximize Ng on the break-in engine without leaving the ground. Do not exceed 100% Ng. █
24. Maintain maximum Ng for 30 seconds.
25. Slowly reduce collective until Ng of break-in engine is 95% ($\pm 1\%$). █
26. Maintain 95% Ng for 30 seconds.
27. Slowly increase collective to maximize Ng on the break-in engine without leaving the ground. Do not exceed 100% Ng. █
28. Maintain maximum Ng for 30 seconds.
29. Slowly decrease collective to full down.
30. Procedure complete.

2.12 ELECTRICAL SYSTEM

2.12.1 AC GENERATOR UNDERFREQUENCY

CAUTION

If generators do not drop out by 90% Nr, return the throttles to FLY. Nr speeds lower than 90% may cause damage to avionics.

1. #1 and #2 GEN – ON.
2. APU – ON.



3. APU GEN switch – ON.
4. AVC – OFF.
5. On EICAS display, select PRESS SHOW OBE bezel. (EICAS will then show HIDE OBE).
6. Place the weather radar in SBY or OFF.
7. Place the transponder in standby.
8. Pull WOW #1 circuit breaker (BATT HOLD UP Bus).
9. Throttles – Retard slowly simultaneously to 90% Nr.
 - a. Verify that the AC 2 GEN FAIL cautions light illuminates between 96% and 92% Nr.
 - b. Verify that the AC 1 GEN FAIL caution light does not illuminate.
 - c. Verify that both LOW ROTOR warning lights illuminate.
 - d. Verify that the "LOW ROTOR" aural alert activates.
10. Throttles – Return to FLY.
11. Push in WOW #1 circuit breaker
12. Pull WOW #2 circuit breaker (BATT Bus).
13. Throttles – Retard slowly simultaneously to 90% Nr.
 - a. Verify that the AC 1 GEN FAIL cautions light illuminates between 96% and 92% Nr.
 - b. Verify that the AC 2 GEN FAIL caution light does not illuminate.
 - c. Verify that both LOW ROTOR warning lights illuminate.
 - d. Verify that the "LOW ROTOR" aural alert activates.
14. Throttles – Return to FLY.
15. Push in WOW #2 circuit breaker.
16. Transponder as required.
17. Weather radar as required.

2.12.2 AC PRIMARY BUS TIE CONNECTOR

1. Select AC GEN NO. 1 – OFF.



- a. Verify that the AC GEN 1 FAIL caution light illuminates.
- b. Verify that the DC CONV 1 FAIL caution light remains extinguished.
2. Select AC GEN NO. 1 – ON.
3. Repeat for #2 AC GENERATOR.

2.12.3 DC BUS TIE CONNECTOR

1. Select DC CONV NO. 1 – OFF.
 - a. Verify that the CONV 1 FAIL caution light illuminates.
 - b. Verify all MFDs function normally.
2. Select DC CONV NO. 1 – ON.
3. Repeat for #2 converter.

2.12.4 BACKUP CONVERTER CHECK



Failure of the backup converter with the #1 and #2 converters OFF will result in loss of collective and cyclic trim. The pilot should guard the controls during the test.

NOTE

The BMS will not be operative with the backup converter supplying DC power. The BMS may need to be reset after this check.

NOTE

With the #1 and #2 converters OFF, the backup converter powers the DC essential and battery buses. In this condition, the battery powers the battery holdup and battery utility buses.



NOTE

If an engine is not running, TGT and engine oil pressure for that engine will not be displayed with the backup converter supplying DC power. In addition the following caution lights for that engine will illuminate: ENG CHIP, FADEC B FAIL, FADEC TLD FAULT, ENG OIL PRES, and BATT CHARGE FAIL.

1. Select DC CONV BACKUP – NORM.
2. Select DC CONV NO. 1 and 2 – OFF.
 - a. Verify that only the pilot inboard and outboard MFDs remain ON.
 - b. Verify that the pilot's AHRS (attitude and heading) remains ON.
 - c. Verify that the pilot's DCU (pilot's engine instruments) remain ON.
 - d. Verify that only the copilot RTU remains ON.
3. Select DC CONV NO. 1 and 2 – ON.

2.13 WINDSHIELD ANTI-ICE



Do not conduct the windshield anti-ice check if OAT is above 26°C.

NOTE

Windshield will not heat if temperature is above 26°C.

1. Place the APU GEN switch to the ON position.
2. Place the AC GEN 1 and 2 switches to the OFF position.
3. Place the ANTI-ICE WDSHLD PLT and CPLT switches to the ON position.
 - a. Verify that neither the pilot nor the copilot windshield heats.
4. Place the ANTI-ICE WDSHLD PLT switch to the OVRD position.
 - a. Verify that only the pilot's windshield heats.
5. Place the ANTI-ICE WDSHLD PLT switch to the OFF position.



6. Place the AC GEN 1 and 2 switches to the ON position.
7. Place the ANTI-ICE WDSHLD PLT and CPLT switches to the ON position.
 - a. Verify that both pilot and copilot windshields heat.
8. Place the ANTI-ICE WDSHLD PLT and CPLT switches to the OFF position.

2.14 AFCS GROUND RUN CHECK

NOTE

A jump in the flight controls or tip path plane could indicate an accelerometer or rate gyro bias. If a jump is noted an ACCEL NULL may be performed through the AFCS MODE SEL panel and TEST.

1. At 105% Nr, select YAW, CYCLIC, and COLL TRIM ON.
2. Select ATT, AP 1, AP 2, SAS 2 PRI and SAS 1 PRI ON.
3. Without engaging any trim releases (pedals, cyclic or collective), select AP 1 OFF.
 - a. Verify that the flight controls do not jump.
 - b. Verify that the rotor tip path plane does not move appreciably.
 - c. Verify that the AFCS stays in ATT Mode.
4. Select AP 1 ON.
5. Select SAS 1 OFF.
 - a. Verify that the flight controls do not jump.
 - b. Verify that the rotor tip path plane does not move appreciably.
 - c. Verify that the AFCS stays in ATT Mode.
6. Select SAS 1 ON.
7. Repeat for AP 2 and SAS 2.



2.15 AUTOMATIC VIBRATION CONTROL

NOTE

The AVC IBIT buzzes each individual force generator.

1. Set Nr at 105% Nr and collective at flat pitch.
2. Place the VIB CTRL switch in the ON position.
3. Select HLTH→AVC.
 - a. Verify that the AVC status is OPERATIONAL.
4. Select IBIT.
 - a. Verify that the AVC status changes to TEST IN PROGRESS.
 - b. Note the buzzing of each force generator.
 - c. Verify that the AVC status changes to TEST PASSED.
 - d. Verify that the AVC status changes to OPERATIONAL.

2.16 GROUND MAIN ROTOR TRACK AND BALANCE

NOTE

Actual track and balance limits are in Chapter 18 in the Maintenance Manual. Limits in this FCP might not be the most current limits and should be used for pilot reference only.

NOTE

The HUMS will continue to average data unless power is removed from the HUMS for 5 minutes. Therefore, between data runs a CREW CHANGE must be initiated, power must be removed from the HUMS for 5 minutes and the data card must be removed and reinstalled. Power can be removed from the HUMS by either pulling the HUMS circuit breaker or shutting down all power on the helicopter.

NOTE

Main rotor ground track must be within limits before commencing ground balance.



NOTE

Tracking space limit only applies to initial blade tracking during ground runs. There is no blade track limit once inflight tuning commences.

2.16.1 GROUND TRACK

1. Select HLTH→HUMS→TRACK BAL→MAIN RTR.
2. Set 105% Nr at flat pitch.
3. Select the GRND RGMES.
4. Select FLAT TRACK.
 - a. When data is displayed, verify that the track spacing is within 20mm.
5. If within limits, continue with ground balancing.
6. If out of limits, select VIEW TRACK.
7. Select FLTTRK SOLN.
8. Note solution and predictions.

2.16.2 GROUND BALANCE

1. Select HLTH→HUMS→TRACK BAL→MAIN RTR.
2. Select AVC ON.
3. Set 105% Nr.
4. Set flat pitch.
5. Select the GRND RGMES soft bezel key.
6. Select GROUND.
 - a. When data is displayed, verify that the helicopter is within limits.
7. Select GRD WT SOLN.
8. Note solution and predictions.

Acceptable Main Rotor Ground 1P Vibration Limits:



	VERT	ROLL	FLR LAT	TRK SPC
Ground	None	.2 IPS	.25 IPS	20mm

9. For subsequent data runs, perform a crew change, remove the HDC, and remove power from the HUMS for 5 minutes.

2.17 TAIL ROTOR BALANCE

NOTE

The HUMS will continue to average data unless power is removed from the HUMS for 5 minutes. Therefore, between data runs a CREW CHANGE must be initiated, power must be removed from the HUMS for 5 minutes and the data card must be removed and reinstalled. Power can be removed from the HUMS by either pulling the HUMS circuit breaker or shutting down all power on the helicopter.

NOTE

The tail rotor is balanced to 0.2 IPS.

NOTE

The tail rotor operational limit is 0.8 IPS.

1. Select HLTH→HUMS→TRACK BAL→TAIL RTR.
2. Select ACQ DATA.
 - a. When data is displayed, verify that the helicopter is within limits.
3. Select BAL SOLN.
4. Note solution and predictions.

Acceptable Tail Rotor Ground 1P Vibration Limits:

T/R Balance	0.2 IPS
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2.18 ENGINE HIGH SPEED SHAFT BALANCE

NOTE

Actual track and balance limits are in Chapter 18 in the Maintenance Manual. Limits in this FCP might not be the most current limits and should be used for pilot reference only.

NOTE

Engine not being checked must be OFF for engine drive shaft check.

NOTE

Engine being checked must drive the drive train exclusively for 5 minutes before data is taken.

NOTE

The high speed shaft is balanced to 0.5 IPS.

NOTE

The high speed shaft operational limit is 1.5 IPS.

NOTE

If vibration level is above 4.0 IPS, the shaft must be indexed prior to another balance.

NOTE

The HUMS will continue to average data unless power is removed from the HUMS for 5 minutes. Therefore, between data runs a CREW CHANGE must be initiated, power must be removed from the HUMS for 5 minutes and the data card must be removed and reinstalled. Power can be removed from the HUMS by either pulling the HUMS circuit breaker or shutting down all power on the helicopter.

1. With one engine running at 105% Nr and flat pitch.
2. Select HLTH→HUMS→ENG TRANS DIAG.
3. Select SHAFT BAL.

2.18.1 NEW BALANCE

1. Select ENG 1 or 2 NEW BALANCE.



2. Select CLEAR ALL WEIGHTS.
 - a. Verify all fields are NO. (XX are not NO).
3. Select CONT.
4. After 30 seconds, select ACQ DATA.
 - a. When the data is displayed, if the vibration data is above 4.0 IPS, shut down engine and have maintenance crew index (rotate) the output shaft to the lowest achievable TIA in accordance with Chapter 18 of the AMM.
 - b. If the vibration data is below 4.0 IPS, continue the run-up.
5. After 5 minutes, select ACQ DATA.
 - a. When data is displayed, verify that the helicopter is within limits.
6. If not within limits, input number of adjustments and select ENTER.
7. Select CALC SOLN.
8. After solution calculations are completed, note predicted amplitude and phase.
9. Select VIEW SOLN.
10. Record solution.

2.18.2 RESUME BALANCE

1. Select ENG 1 or 2 RESUME BALANCE.
 - a. Verify that the weights in all the fields correspond to the actual weights on the shaft. If not, use instructions on screen to correct.
2. Select CONT.
3. After 5 minutes, select ACQ DATA.
 - a. When data is displayed, verify that the helicopter is within limits.
4. If not within limits, input number of adjustments and select ENTER.
5. Select CALC SOLN.
6. After solution calculations are completed, note predicted amplitude and phase.
7. Select VIEW SOLN.



8. Record solution.

Acceptable High Speed Shaft Balance Vibration Limit

Engine Drive Shafts 0.5 IPS

2.19 ENGINE HIGH SPEED SHAFT DIAGNOSTIC

NOTE

This procedure can be used to verify high speed shaft operational limits without removing the engine inlets.

NOTE

Engine not being checked must be at ground IDLE or OFF for Engine Drive Shaft check.

NOTE

Engine being checked must drive the drive train exclusively for 5 minutes before data is taken (other engine at 0% Q).

NOTE

No balance solution is provided.

NOTE

The high speed shaft operational limit is 1.5 IPS.

1. Set the #2 engine at ground IDLE or OFF.
2. With #1 engine running at 105% Nr and flat pitch, wait 5 minutes.
3. Select HLTH→HUMS→ENG TRANS DIAG→LEFT MOD MD.
4. Select FORCE.
5. After computations are complete and displayed, verify that the #1 Engine high speed shaft is within limits.
6. Repeat for #2 Engine using the RIGHT MOD MD.



2.20 ENGINE ANTI-ICE

NOTE

The TRN INLT A-1 ON caution light will illuminate when the OAT is less than 41°F and either of the engine anti-ice switches are in the OFF position.

1. Select ANTI-ICE - ENGINE 1 switch – ON.
 - a. Verify that the INLT 1 A-ICE advisory light illuminates.
 - b. Verify that the INLT 1 A-ICE FAIL caution light is extinguished.
 - c. Verify that the ENG 1 A-ICE DGRD caution light is extinguished.
2. Select ANTI-ICE - ENGINE 1 switch – OFF.
3. Repeat for #2 engine.

2.21 FUEL SELECTOR AND SHUTOFF VALVE



Engine shutdowns shall be performed from the IDLE position.

NOTE

Each engine will be shut down twice; once with the main fuel valve switch and another with FIRE/ARM pushbutton. These switches both activate the main fuel selector/shutoff valve at the engine fire wall.

NOTE

From IDLE, allow approximately 20 seconds for the engine to shut down.

NOTE

The APU shuts down almost immediately upon depressing the FIRE/ARM pushbutton.



NOTE

The FADEC may generate an ENG OVSP TEST caution light when an engine is shut down using the fuel selector or fuel shutoff valve. The FADEC will reset the caution light after two successive engine starts.

2.21.1 MAIN FUEL SHUTOFF

1. Place the NO. 1 ENG throttle – IDLE.
2. Select NO. 1 FUEL – OFF.
 - a. Verify that the #1 engine green fuel indication line on the EICAS disappears.
 - b. Verify that the #1 engine shuts down from fuel starvation.
3. Select NO. 1 ENG throttle – OFF.
4. Repeat for NO. 2 engine.

2.21.2 FIRE/ARM SWITCHES

1. Place the NO. 1 ENG throttle – IDLE.
2. Press the NO. 1 ENG FIRE/ARM button.
 - a. Verify that the ARM light illuminates.
 - b. Verify that the #1 engine shuts down from fuel starvation.
3. Place NO.1 ENG Throttle – STOP.
4. Deselect the NO. 1 ENG FIRE/ARM button.
 - a. Verify that the ARM light extinguishes.
5. Repeat for NO. 2 Engine.
6. Select APU GEN – OFF.
7. Press APU FIRE pushbutton.
 - a. Verify that the ARM light illuminates.
 - b. Verify that the APU shuts down from fuel starvation immediately.
8. Select APU CONT switch – OFF.



9. Deselect APU FIRE pushbutton.
 - a. Verify that the ARM light extinguishes.

2.22 DOOR

1. Have crew close the cabin door.
 - a. Verify that the DOOR OPEN advisory light extinguishes.
2. Have the crew open the cabin door.
 - a. Verify that the DOOR OPEN advisory light illuminates.
3. With the cabin door closed, have the crew open the external power door.
 - a. Verify that the DOOR OPEN advisory light illuminates.
4. Have the crew close the external power door.
 - a. Verify that the DOOR OPEN advisory light extinguishes.

2.23 RAMP



Ensure ramp area is clear before actuating ramp.

1. Have the crew open the upper ramp door with the lower ramp door closed.
 - a. Verify that the RAMP OPEN advisory light illuminates.
2. Have the crew close the upper ramp door.
 - a. Verify that the RAMP OPEN advisory light extinguishes.
3. Have the crew open and close the lower ramp door with the upper door closed using the cabin ramp control switches.
 - a. Verify that the RAMP OPEN advisory light illuminated when the ramp is open.
 - b. Verify that the RAMP OPEN advisory light extinguishes when the ramp is closed.
4. Open and close the lower ramp door using the cockpit control switches.



- a. Verify that the RAMP OPEN advisory light illuminated when the ramp is open.
- b. Verify that the RAMP OPEN advisory light extinguishes when the ramp is closed.

2.24 ROTOR BRAKE BREAK-IN

NOTE

The purpose of the rotor brake break-in procedure is to ensure proper mating of components. It is required when components (rotor brake disc, pucks or housing) are changed or serviced.

1. Shut down both engines.
2. As Nr decreases to below 40%, apply the rotor brake using a pressure of 150 – 180 psi.
3. After a 15-minute cool-down period, repeat this procedure two additional times to complete the break-in cycle.
4. After break-in cycle is completed:
 - a. Verify that the rotor brake will hold with both engines at IDLE.

2.25 MAINTENANCE DATA COMPUTER

NOTE

The status of a system that is not OPERATIONAL can be NOT CONFIGURED, OFF, INOPERATIVE, MAINTENANCE, NO COMMUNICATION, or WEAK SIGNAL. Only one FUEL PUMP is operative at any one time.

1. Verify that the following systems are OPERATIONAL by selecting HLTH on any MFD:

HYDRAULICS

RIPS (if installed)

AFCS

AFCS 1

AFCS 2

AVC

AVIONICS



ELECTRICAL

DRIVE TRAIN

PROPULSION

AIRFRAME

FUEL

PUMP 1

PUMP 2

HUMS

2. If a system cannot be made OPERATIONAL, select SUB STAT and note the status and the fault code of the inoperative subsystem.

2.26 GROUND ENGINE POWER ASSURANCE CHECK

NOTE

The EPAC history data is stored in the DCU. Each DCU stores historical EPAC data for both engines.

NOTE

After replacing an engine, use the administration functions on the EPAC page to clear data from the previous engine. After replacing a DCU, use the administration functions to cross fill data from the other DCU.

NOTE

The EPAC administrative functions are only available on APU or external power prior to starting the engines.

NOTE

The XFILL function on the EPAC display should be used when a DCU is replaced. If DCU 2 is replaced, on the ground activate XFILL 1 TO 2. If the DCUs are swapped for troubleshooting, no XFILL action is required. If a FADEC is replaced or the FADECs are swapped, no XFILL action is required.

NOTE

If an engine is replaced, the CLEAR ENG and CONFIRM CLEAR functions should be used.



NOTE

Do not replace both DCUs at the same time, EPAC data will be lost.
Replace one at a time and XFILL.

NOTE

A light helicopter and/or a maladjusted WOW switch might initiate an
In Flight EPAC instead of a Ground EPAC.

NOTE

The minimum TGT margin for a new engine is 30°C.

NOTE

The TGT margin for a removed and reinstalled engine should be the
same margin as before the engine was removed.

1. Position the helicopter into the wind.
2. Select CABIN and COCKPIT HEATER – OFF.
3. Select UTIL hard bezel on an MFD.
4. Select EPAC soft bezel.
5. Set the NO. 2 Throttle to IDLE (0% Torque).
6. Set the NO. 1 Throttle to FLY at 105% Np.
7. Increase collective to 50% torque.
8. Select INIT #1.
9. Upon completion of test, select ACCEPT ENG 1 if PASSED (See Note).
 - a. Verify data logged on EPAC page.
10. Set NO. 2 Throttle to FLY.
11. Set NO. 1 Throttle to IDLE.
 - a. Verify that torque is within 6% of the original setting (50%).
12. Adjust collective to 50% torque.
13. Select INIT #2.
14. Upon completion of test, select ACCEPT ENG 2 if PASSED (See Note).



- a. Verify data logged on EPAC page.
- 15. Reduce collective to flat pitch.
- 16. Set the NO. 1 throttle to FLY.

2.27 FADEC TRAINING MODE

- 1. With both engines in FLY, set collective at Flat Pitch.
- 2. Note torque and TGT for each engine.
- 3. Without moving the collective, on the OEI TRAIN panel place the Dual Training switch to DUAL TRAIN.
 - a. Verify that TRAIN is displayed with a blue background on the PFD.
 - b. Verify that TRAIN is displayed with a blue background on the EICAS over both engine pods.
 - c. Verify that both torques increase approximately 15%.
 - d. Verify that both TGTs increase approximately 35° - 110°.
- 4. Place the TRNG IDLE switch to #1.
 - a. Verify that the #1 engine Np drops to approximately 92 - 95%.
 - b. Verify that the #1 engine torque drops.
 - c. Verify on the PFD that the #2 engine 30S TGT is armed.
 - d. Verify that the Nr green arc changes to 100 - 106%.
- 5. Using the pilot collective OEI switch, select 2 minute power by pushing aft on the switch.
 - a. Verify on the PFD that the #2 engine 2M TGT is armed.
- 6. Using the pilot collective OEI switch, select maximum continuous power by pushing down on the switch.
 - a. Verify on the PFD that the #2 engine indicates nothing is armed (Max Continuous).
- 7. Select TRNG IDLE.
 - a. Verify that both engines regain training power.
- 8. Place the TRNG IDLE switch to #2.
 - a. Verify that neither engine loses power.
- 9. Select TRNG IDLE.



10. Select DUAL TRAIN switch to OFF.
11. Repeat for #2 engine and the copilot OEI collective switch.

2.28 BMS

NOTE

Oil cooler and hangar bearings temperature/vibration exceedances, as well as swashplate vibration exceedances, only activate the CHECK BMS advisory.

NOTE

The CHECK BMS advisory is not initially displayed in the air (i.e., it is not displayed until landing).

NOTE

Swashplate temperature exceedances activate the S/P TEMP caution.

1. Select the HLTH status page.
2. Select DRIVE TRAIN→ SUB SYS→X BEARING.
3. Select XCEED (if displayed).
4. Record any exceedance.

2.29 HUMS

NOTE

HUMS exceedances will automatically be displayed on the HLTH page after touchdown.

NOTE

Active exceedances are listed on the top of the Exceedance page. Details of each exceedance such as duration and maximum/minimum value are listed at the bottom of the page for the selected exceedance.

NOTE

The most common cause of a COLD START exceedance is the HUMS being powered down without a CREW CHANGE on the previous flight.



1. To view exceedances, select HLTH and HUMS.
2. Select ACTIVE EXCD.
3. Select exceedance and record details of any Exceedance Message except COLD START.

2.30 WHEEL BRAKE BREAK-IN

NOTE

The purpose of the wheel brake break-in procedure is to ensure proper mating of components.

NOTE

Wait 5 minutes after last application of brakes before applying the parking brake.

1. On a flat surface, ground taxi between 8 to 10 knots ground speed.
2. Lower the collective.
3. Start timing.
4. From the pilot position, very slowly apply brakes evenly and steadily to come to a complete stop in 15 to 20 seconds.
 - a. Verify that the brake pressure is even between the right and left pedals.
 - b. Verify that the brake system efficiently stops the helicopter.
5. Repeat three more times.
6. From the copilot's position, apply both breaks.
 - a. Verify that the brake pressure is even between the right and left pedals.
 - b. Verify that the brake system efficiently stops the helicopter.



3.0 HOVER CHECKS

3.1 FLIGHT MAIN ROTOR TRACK

NOTE

Actual 1P, 2P and 3P track and balance limits are in Chapter 18 of the S-92 Maintenance Manual. Limits in this FCP might not be the most current limits and should be used for pilot reference only.

NOTE

The HUMS will continue to take and process data anytime the flight conditions are satisfied.

NOTE

2P and 3P data is taken in the background during the 1P acquisition and only accessible through the ground station.

NOTE

2P corrections can be included in the 1P corrections if selected during ground station rotor tuning setup – see Chapter 18 of the S-92 Maintenance Manual.

NOTE

If a 3P limit is exceeded, contact Sikorsky Engineering for a correction.

NOTE

The aircraft is limited to 86% torque above 100 KIAS. If 150 KIAS cannot be achieved, set 86% torque and accept the airspeed to take data.

3.1.1 MAIN ROTOR 1P, 2P AND 3P

1. Select HLTH→HUMS→TRACK BAL→MAIN RTR.
2. Select AVC ON.
3. Set flat pitch with the helicopter on the ground.
4. Select the GROUND soft bezel key.



- a. When data is displayed, verify that the helicopter is within limits.

Acceptable Main Rotor Ground 1P Vibration Limits:

	VERT	ROLL	FLR LAT	TRK SPC
Ground	None	.2 IPS	.25 IPS	None

5. Hover the helicopter.

6. Select FLT RGMES.

7. Select HOVER.

- a. When data is displayed, verify that the helicopter is within limits.

8. Fly helicopter straight and level.

9. Repeat for 80 KIAS, 120 KIAS, 140 KIAS and 150 KIAS.

Acceptable Main Rotor Flight 1P Vibration Limits:

	VERT	OH LAT	TRK SPC
Hover	.2	.25	None
80 KIAS	.2	.25	None
120 KIAS	.2	.25	None
140 KIAS	.2	.25	None
150 KIAS	.2	.25	None

Acceptable Main Rotor Flight 2P and 3P Vibration Limits:

	2P Cockpit Vertical	3P Cockpit Vertical
Hover	.2	.2
80 KIAS	.2	.2
120 KIAS	.2	.2
140 KIAS	.2	.2
150 KIAS	.2	.2

3.2 WEIGHT ON WHEELS CHECK

NOTE

The flight time recorder will not indicate flight time until after 1 minute airborne.

NOTE

The right WOW switch controls the clock and weather radar.



NOTE

The left WOW switch controls the FMS 1 and Transponder 1.

1. Select WX on the weather radar.
2. Select WX overlay on the NAV display.
 - a. Verify that the weather radar is in FSTBY.
3. Select FT on the clock.
4. Select DATA 2/4 on the FMS.
 - a. Verify that the VOR, ADC, and DME are deselected (D).
5. Lift helicopter into a hover.
 - a. Verify that the WX display changes from the FSTBY to WX.
 - b. Verify that the flight time counter starts recording flight time.
 - c. Verify that the VOR, ADC, and DME switch from DESEL to SELECT on DATA 2/4 on the FMS.
 - d. Verify that the transient Nr droop is 1% or less.

3.3 INITIAL HOVER CHECKS

NOTE

Momentary oil pressure fluctuations may occur during transient maneuvers (e.g., pitch low takeoffs, hover checks, gusty wind conditions). These types of fluctuations are acceptable even when oil pressure drops into the yellow range (below 45 psi).

1. Hover helicopter into the wind.
 - a. Verify that Nr is set to 105% Nr.
 - b. Verify that the helicopter can be controlled in pitch, roll, yaw and altitude.
 - c. Verify that the hydraulic, main gear box and engine instruments on the EICAS page are in the green.
 - d. During steady state hover conditions, verify that the MGB oil pressure is between 55 and 60 psi.
 - e. Verify that the aircraft exhibits no unusual vibration.
 - f. Verify that the #1 and #2 torques are matched within 5%.



- g. Verify that both airspeed indicators on PFD and backup airspeed indicator read zero plus wind speed.

3.4 HOVER CONTROL POSITION CHECK

NOTE

For reference, the large nuts securing the pedals are 1.3125" from flat to flat.

NOTE

The autorotation Nr must be within limits to get a valid pedal position check. Do not continue with the flight checks past the Autorotation Check unless the Hover Control Position Check is within limits as described below.

NOTE

This pedal check gives the pilot an indication the tail rotor is safely rigged. It is not intended as a precise rigging check.

1. Hover the helicopter into the wind.
 - a. Verify that the cyclic position is approximately 1" forward of neutral.
 - b. Verify that the left pedal is $1\frac{1}{2}" \pm \frac{3}{4}"$ forward of the right pedal.

3.5 COMPASS AND HSI CHECK

NOTE

The FAA requires that standby compass calibration correction be less than 10° from actual magnetic heading.

1. Initiate a pedal turn stopping at the four cardinal (N, E, S and W) headings.
 - a. Verify the smooth operation of pilot and copilot HSI.
 - b. Verify the smooth operation of the standby compass.
2. Compare standby magnetic compass with the HSI.
 - a. Note the compass corrections on the compass calibration card.
 - b. Verify that the standby compass is within $\pm 10^\circ$ of the actual magnetic heading.



3.6 AFCS HOVER CHECKS

3.6.1 SAS

NOTE

Control input should be crisp, allowing the cyclic to return to the trimmed position in a controlled/guarded manner. Input rate should be such that the desired attitude change is obtained and the cyclic returned to the trimmed position in approximately one second.

NOTE

A properly functioning PSAS/SSAS will dampen the induced pitch/roll rate but will not necessarily return the helicopter to the trimmed attitude. Therefore, the tendency of the helicopter to stop pitching/rolling during single SAS checks rather than pitch/roll back to the original attitude confirms that PSAS/SSAS is functioning properly.

NOTE

If helicopter loading/configuration results in a left wing down hover attitude of approximately 2° or more, the roll proportional stability contribution of the SAS will be saturated. However, the roll rate damping function of the SAS should still function properly. Consequently, the helicopter should exhibit a damped response to a left control input but will not exhibit a tendency to return to trim. The tendency to return to trim should not be confused with a damped roll response.

1. Select YAW, CYCLIC and COLL TRIM – ON.
2. #1 PSAS Check:
 - a. Select SAS 1 and SAS 1 PRI – ON.
 - b. Select AP2, AP1 and SAS 2 – OFF.
 - c. Verify that the SAS 2 PRES caution light illuminates.
 - d. Introduce a forward 3° pitch attitude pulse (push cyclic forward and release). Verify that the pitch rate is damped (slows down or stops).
 - e. Repeat using an aft 3° pitch attitude pulse.
 - f. Repeat using a left 5° roll attitude pulse.



- g. Repeat using a right 5° roll attitude pulse.
3. #2 PSAS Check:
- a. Select SAS 2 and SAS 2 PRI – ON.
 - b. Select AP1, AP2 and SAS 1 – OFF.
 - c. Verify that the SAS 1 PRES caution light illuminates.
 - d. Introduce a forward 3° pitch attitude pulse (push cyclic forward and release). Verify that the pitch rate is damped (slows down or stops).
 - e. Repeat using an aft 3° pitch attitude pulse.
 - f. Repeat using a left 5° roll attitude pulse.
 - g. Repeat using a right 5° roll attitude pulse.
4. #1 SSAS Check:
- a. Select SAS 1 and SAS 1 SEC – ON.
 - b. Select AP2, AP1 and SAS2 – OFF.
 - c. Verify that the SAS 2 PRES caution light illuminates.
 - d. Introduce a forward 3° pitch attitude pulse (push cyclic forward and release). Verify that the pitch rate is damped (slows down or stops).
 - e. Repeat using an aft 3° pitch attitude pulse.
 - f. Repeat using a left 5° roll attitude pulse.
 - g. Repeat using a right 5° roll attitude pulse.
5. #2 SSAS Check:
- a. Select SAS 2 and SAS2 SEC – ON.
 - b. Select AP1, AP2 and SAS1 – OFF.
 - c. Verify that the SAS 1 PRES caution light illuminates.
 - d. Introduce a forward 3° pitch attitude pulse (push cyclic forward and release).
 - e. Verify that the pitch rate is damped (slows down or stops).
 - f. Repeat using an aft 3° pitch attitude pulse.
 - g. Repeat using a left 5° roll attitude pulse.



- h. Repeat using a right 5° roll attitude pulse.
- 6. Select ATT, AP 1, AP 2, SAS 1 PRI, and SAS 2 PRI – ON.

3.6.2 AUTOPILOT

NOTE

Control input should be crisp, allowing the cyclic to return to the trimmed position in a controlled/guarded manner. Input rate should be such that the desired attitude change is obtained and the cyclic returned to the trimmed position in approximately one second.

- 1. Select YAW, CYCLIC and COLL TRIM – ON.
- 2. Select ATT, AP 1, SAS 1, and PRI – ON.
- 3. Select SAS 2 and PRI – ON.
- 4. Select AP 2 – OFF.
- 5. In a steady state hover:
 - a. Verify that the hands-off attitude retention is $\pm 1^\circ$.
 - b. Verify that the heading hold is within $\pm 2^\circ$.
- 6. Introduce a forward 5° pitch attitude pulse (push cyclic forward and release).
 - a. Verify that the helicopter returns to the trimmed attitude with no more than one overshoot.
 - b. Verify that the cyclic moves in response to the inputs to return the helicopter to a trimmed attitude.
- 7. Repeat using an aft 5° pitch attitude pulse.
- 8. Introduce a left 10° roll attitude pulse (push cyclic left and release).
 - a. Verify that the helicopter returns to the trimmed attitude with no more than one overshoot.
 - b. Verify that the cyclic moves in response to the inputs to return the helicopter to a trimmed attitude.
- 9. Repeat using a right 10° roll attitude pulse.
- 10. Make a 20% TRQ change with feet off the pedals.
 - a. Verify that the helicopter maintains $\pm 2^\circ$ of initial heading (heading hold).
 - b. Verify that the pedals move during TRQ change to maintain heading.



11. Select AP 2 – ON.
12. Select AP 1 – OFF.
13. Repeat checks for AP 2.

3.6.3 RADAR ALTITUDE HOLD

1. YAW, CYCLIC, and COLL TRIM – ON.
2. ATT, AP 1, SAS 1, and PRI – ON.
3. AP 2, SAS 2, and PRI – ON.
4. Select FD 2 on the AFCS control panel.
5. Hover helicopter at an altitude between 50 and 75 feet (selected altitude).
6. Select RALT on the pilot FD MODE SEL panel and confirm the following:
 - a. The COUPLE light illuminates on the AFCS control panel.
 - b. CPL is displayed in green under FD 2 on the PFD.
 - c. RALT ON is displayed on both MODE SEL panels.
 - d. RALT is displayed in green on top of the PFD by the C label (collective channel).
 - e. The selected radar altitude is displayed in magenta over the VSI and a magenta bug appears on the radar altimeter display (for helicopters equipped with AMS 4.1).
 - f. The helicopter maintains the selected altitude ± 4 feet.
7. Without depressing the collective trim, push collective down to descend 5 feet and release collective. Helicopter should return to selected altitude within ± 4 feet and with no more than a 4-foot overshoot.
8. Depress the collective trim and slowly change the hover altitude by at least 10 feet.
9. Releasing the collective trim, the helicopter should maintain the altitude at which the trim was released within ± 4 feet.
10. Use the pilot collective beeper to change altitude by at least 10 feet.
11. Change the helicopter heading by 5° using the pilot's collective HDG TRIM switch.
12. Depress the pilot cyclic Flight Director Standby button to decouple FD 2.
 - a. Verify that FD 2 decouples.



- b. Verify that the "DECOUPLE" aural alert is heard.
13. Select FD 1 on the AFCS control panel.
14. Repeat on the copilot's side using the copilot's PFD, HDG TRIM switch, cyclic FD standby button, collective beeper and MODE SEL panel.



4.0 FLIGHT CHECKS

4.1 FLIGHT INSTRUMENTS

1. Check the pilot, copilot and backup airspeed indicators.
 - a. Maximum allowable difference among pilot, copilot and backup airspeed indicators is ± 5 knots.
 - b. Maximum allowable fluctuation per indicator in straight and level flight is ± 3 knots.
2. Check the pilot HSI, copilot HSI, and standby compass for smooth operation and agreement.
3. Check the standby attitude indicator for smooth operation and agreement with the PDFs.
4. Check the VSI and altimeters for smooth operation.

4.2 AUTOROTATION

WARNING

If an Nr of at least 90% cannot be maintained, discontinue the autorotation and land.

CAUTION

Do not allow Nr to exceed 110% or go below 90% Nr during this check.

CAUTION

Do not exceed 106.2% Np for more than 12 seconds.

NOTE

To reduce the potential for Np or Nr overspeeds, select a collective setting (10%, 15%, or 20%) that will result in an Nr below 105% at the test AGW and density altitude. This will require retarding the throttles to achieve an autorotation state.



4.2.1 GROUND RUN (ROTORS NOT TURNING)

1. Select HLTH→HUMS→MAINT→PARAMS→ENTR→ANLG→ANLG2.
2. Raise collective full up.
 - a. Record collective position.
3. Lower collective full down.
 - a. Record collective position.
4. Determine the collective position number that corresponds to 10%, 15%, and 20% from the collective bottom by using the numbers determined during the ground run.

EXAMPLE:

Collective Up	55
Collective Down	-60
Range	115
10% (-60+11.5)	-48.5
15% (-60+17.25)	-42.75
20% (-60+23)	-37.0

4.2.2 FLIGHT CHECK

1. Set barometric altimeter to 29.92.
2. Select HLTH→HUMS→MAINT→PARAMS→ENTR→ANLG→ANLG2.
3. Lower collective to establish 80 KIAS autorotation (airspeed hold may be used).
 - a. Verify that Np 1 and Np 2 separate from Nr.
4. Retard throttles to obtain Np/Nr split, if required.
5. Set collective at 10%, 15%, or 20% to establish Nr between 90 – 110% Nr.
6. At check altitude, record Nr.
7. Return throttles to FLY.
8. Raise collective to recover from the autorotation.
9. Record OAT at check altitude.
10. Record aircraft gross weight, check altitude and collective setting used.



11. Calculate density altitude using pressure altitude (check altitude at 29.92), OAT, and Figure 1 in this FCP.
12. Using density altitude and aircraft gross weight, refer to the 10%, 15%, or 20% Autorotation Charts in this FCP to calculate acceptable Nr.
13. The maximum allowable difference between actual and chart Nr value is $\pm 3\%$.

4.3 LANDING GEAR CHECK

1. Place the LDG GEAR lever in the UP position.
 - a. Verify that the landing gear retracts.
 - b. Verify that the landing gear position lights extinguish.
2. Place the LDG GEAR lever in the DN position.
 - a. Verify that the landing gear extends.
 - b. Verify that the landing gear position lights illuminate green.

4.4 FLIGHT ENGINE POWER ASSURANCE CHECK

NOTE

TGT must be steady to avoid a FADEC ABORT. Hand flying might be steadier than coupled flight.

NOTE

The minimum TGT margin for a new engine is 30°C.

NOTE

The TGT margin for a removed and reinstalled engine should be the same margin as before the engine was removed.

1. Select CABIN and COCKPIT HEATER – OFF.
2. Select UTIL hard bezel on a MFD.
3. Select EPAC soft bezel.
4. Establish level flight at 120 TAS.
5. Select INIT #1.



6. Upon completion of test, select ACCEPT ENG 1 if PASSED (See Note).
 - a. Verify data logged on EPAC page.
7. Select INIT #2.
8. Upon completion of test, select ACCEPT ENG 2 if PASSED (See Note).
 - a. Verify data logged on EPAC page.

4.5 AFCS FLIGHT CHECKS

NOTE

AFCS checks should be performed in the smoothest possible flight conditions.

4.5.1 AUTOPILOT – UNCOUPLED

1. YAW, CYCLIC, and COLL TRIM – ON.
2. ATT, AP 2, SAS 2, and PRI – ON.
3. SAS 1 and PRI – ON.
4. AP 1 – OFF.
5. Establish 120 KIAS straight and level.
6. The steady state attitude/heading retention tolerances in smooth air is $\pm 1^\circ$ in pitch, roll, and heading.
7. Repeat with AP 1 – ON and AP 2 – OFF.

4.5.2 AUTOPILOT – 2 CUE

4.5.2.1 STEEP TURNS

1. YAW, CYCLIC, and COLL TRIM – ON.
2. ATT, AP2, SAS2, and PRI – ON.
3. SAS1 and PRI – ON.
4. AP 1 – OFF.



5. Establish 120 KIAS straight and level.
6. Select FD 2 on the AFCS control panel.
7. Select ALT and IAS on the pilot's MODE SEL panel.
8. With feet off the pedals, roll the helicopter into a right 30° turn using the pilot's cyclic beeper trim.
9. Make a 180° turn and note:
 - a. The ball should stay within ½ ball width of the center during the turn.
 - b. The altitude should stay within ± 50 ft.
 - c. The airspeed should stay within ± 5 KIAS.
10. Using pilot's cyclic trim, roll the helicopter to 0°.
11. With feet off the pedals, roll the helicopter into a left 30° turn against trim.
12. Make a 180° turn and note:
 - a. The ball should stay within ½ ball width of the center during the turn.
 - b. The altitude should stay within ± 50 ft.
 - c. The airspeed should stay within ± 5 KIAS.
13. Roll the helicopter to 0°.
14. Decouple the autopilot by selecting STBY on the pilot's MODE SEL panel; note the "DECOUPLE" aural alert.
15. Select AP 1 – ON and AP 2 – OFF.
16. Select FD1 on the AFCS control panel.
17. Repeat using copilot's cyclic beeper trim and the copilot's MODE SEL panel.

4.5.3 VOR TRACKING

1. Position the helicopter six to ten nautical miles from a VOR on a low altitude airway above the minimum enroute altitude.
2. Input VOR frequency into VOR 2.
3. Select VOR 2 as a navigation source on the pilot's HSI.
4. Select YAW, CYCLIC, and COLL TRIM – ON.



5. Select ATT, AP 1, AP 2, SAS 1 and SAS 2, and PRIs – ON.
6. Establish 120 KIAS straight and level.
7. Select FD 2 on the AFCS control panel.
8. Press HDG PUSH TO SYNC on the remote instrument controller.
9. Select HDG and ALT on the pilots MODE SEL panel.
10. Set COURSE 2 to a radial 15° from the present radial.
11. With hands and feet off the controls, select a heading to intercept the course by rotating the heading control knob at least 45°.
 - a. Verify that the helicopter smoothly rolls into and out of the turn with no more than 2° of overshoot from the selected heading.
 - b. Verify the helicopter altitude retention is ± 50 ft. from the selected altitude.
 - c. The roll attitude does not exceed 20°.
12. Select NAV on the pilot's MODE SEL panel:
 - a. Verify that NAV ARM is displayed on the MSP.
 - b. Verify that the helicopter executes an intercept when the CDI becomes active.
 - c. Verify that the intercept occurs with no more than one overshoot and no more than 20° angle of bank.
 - d. Verify that NAV CAP is displayed when the course is captured.
13. Track the inbound radial.
 - a. Verify that there is no lateral deviation greater than 1/3 dot.
 - b. Verify that the helicopter tracks smoothly.
14. Track the inbound radial over the VOR.
 - a. Verify that the helicopter tracks through station passage and transitions to the outbound radial.
15. Repeat using VOR 1, FD1, copilot's MODE SEL panel and the COURSE 1.

4.5.4 ALTITUDE MODE

1. YAW, CYCLIC and COLL TRIM – ON.
2. ATT, AP 1, AP 2, SAS 1 and SAS 2, and PRIs – ON.



3. Establish 120 KIAS straight and level.
4. Engage HDG and ALT modes on the FD MODE SEL panel.
 - a. Confirm HDG and ALT ON light appear on the FD MODE SEL panel.
 - b. Verify that the altitude holds within ± 50 feet at selected altitude.
5. On the DCP, rotate the REF SET knob to ALTP. Rotate the inner knob of the REF SET to select an altitude 500 feet above current altitude.
 - a. Confirm the selected altitude appears above the BARALT on the pilot PFD in white.
6. Engage ALTP on the pilot FD MODE SEL panel. Confirm the following:
 - a. ALT light on the FD MODE SEL panel extinguishes.
 - b. ALTP ARM appears on the pilot FD MODE SEL panel.
 - c. VS ON is displayed on the pilot FD MODE SEL panel.
 - d. VS displayed in green on the pilot and copilot PFD next to the P label.
 - e. The selected climb rate is displayed both digitally and via a bug on the VSI.
7. Verify that the helicopter immediately starts a 500 fpm climb.
8. Confirm that vertical speed can be adjusted using the cyclic beeper trim.
 - a. The selected climb rate changes on the VSI.
9. At 300 feet prior to the preset altitude confirm:
 - a. Aural alert sounds "LEVEL OFF."
10. Approximately 100 feet prior to selected altitude (proportional to rate of climb or descent) confirm:
 - a. ALTP ARM changes to ALTP CAP.
 - b. VS ON light extinguishes.
 - c. VS on pilot PFD is replaced by ALTP.
11. When altitude captures confirm:
 - a. ALTP CAP light extinguishes.
 - b. ALT ON light on pilot FD MODE SEL panel.
 - c. ALT is displayed on pilot PFD.



12. Verify that the helicopter levels off within 50 feet of the selected altitude with no more than one overshoot.
13. Repeat using FD1 and copilot's FD MODE SEL panel.



4.5.5 AUTOPILOT – 3 CUE

4.5.5.1 ILS

NOTE

Course intercept should occur with no more than one overshoot and a maximum 22° commanded angle of bank at any point.

1. Position the helicopter six to ten nautical miles from an ILS station.
2. Select FD2 on the AFCS control panel.
3. Select NAV2, MIN and RA on the pilot display control panel.
4. Set a COURSE 2 and HEADING to intercept the ILS.
5. Engage IAS, ALT and HDG on the pilot FD MODE SEL panel.
 - a. Confirm proper indications on both FD MODE SEL panels and PFDs.

NOTE

LOC must be armed or captured before selecting GS. GS must be armed or captured before selecting DCL. GS must be captured from below the GS. Adjusting airspeed after activating DCL will cancel that function and requires re-arming of DCL mode.

6. Select the APP hard bezel key on the FD MODE SEL panel.
 - a. Confirm ILS soft bezel key appears on the FD MODE SEL panel.
 - b. Select ILS soft bezel key on the FD MODE SEL panel.
 - c. Confirm APP ILS is displayed on the FD MODE SEL panel.
7. Select the LOC, GS and DCL soft bezel keys.
 - a. Confirm ARM appears in white for each mode (LOC, GS, DCL) on the FD MODE SEL panel.
 - b. Confirm that DCL, LOC and G/S are displayed in white under the P, R and C cues on the pilot PFD.
8. The helicopter will capture and fly the ILS. Confirm the following:
 - a. Smooth capture of the localizer with no more than a ½ dot overshoot.
 - b. FD MODE SEL panel LOC turns from white ARM to green CAP.
 - c. HDG light on FD MODE SEL panel extinguishes.



- d. HDG on PFD is replaced with green LOC.
- e. The course deviation indicator on the HSI and the localizer indicator on the attitude indicator agree.
- f. Marker beacon audio heard concurrently with OM displayed in blue on the PFD.
- g. Smooth capture of the glideslope with no more than a ½ dot overshoot.
- h. FD MODE SEL panel GS turns from white ARM to green CAP.
- i. ALT light on FD MODE SEL panel extinguishes.
- j. ALT on PFD is replaced with green G/S.
- k. Deceleration from selected speed to 60 KIAS at 0.5 kts/sec.
- l. FD MODE SEL panel DCL turns from white ARM to green CAP.
- m. IAS on PFD is replaced with green DCL.
- n. "MINIMUMS" aural alert heard concurrently with MIN displayed in yellow on the PFD at the selected altitude ("EGPWS" aural altitude alert).
- o. "ALTITUDE" aural alert heard at the selected RA altitude ("EGPWS" aural altitude alert).

NOTE

Auto level off and deceleration will not function unless the radar altimeter is on and providing a valid signal.

- p. Helicopter levels off at 50 feet AGL while maintaining localizer and 60 KIAS. DCL is replaced with IAS on the PFD and MSP.
9. Select Go Around mode by pressing the GA button on the pilot cyclic. Confirm the following:
- a. Helicopter transitions to an 80 KIAS, 750 fpm climb on the last heading used to maintain localizer.
 - b. GA ON is displayed in green on both FD MODE SEL panel.
 - c. The pilot may then select HDG, IAS and VS or ALT at the desired values during GA.
10. Decouple by pressing the FD STBY button on the pilot cyclic.
- a. Confirm the "DECOUPLE" aural alert is heard.

4.5.5.2 P-ILS

- 1. Position the helicopter to conduct an FMS approach to an airport with either a GPS or ILS approach.
- 2. Select FD1 on the AFCS control panel.



3. Create a flight plan on the FMS and add the desired FMS approach.
4. Select OVLY and then FPLN on the NAV display.
 - a. Verify that the FMS flight plan is overlaid on the NAV display.
5. Using the LNAV select key on the copilot DCP, select the FMS as the NAV source on the copilot PFD.
6. Set HEADING to intercept the flight plan track.
7. Engage IAS, ALT and HDG on the FD MODE SEL panel.
8. Confirm proper indications on the FD MODE SEL panel and PFD.
9. Select NAV on the copilot FD MODE SEL panel.
 - a. Confirm LNAV appears in green, replacing HDG on the FD MODE SEL panels.
10. Select P-ILS in a preview mode on the copilot PFD using the DCP.
 - a. Confirm P-ILS appears in white over the FMS selection.
 - b. Confirm P-ILS course is displayed in white dashed lines behind the selected FMS course on the copilot PFD.
11. Select the APP hard bezel key on the copilot FD MODE SEL panel.
 - a. Confirm ILS soft bezel key appears on the FD MODE SEL panel.
12. Select ILS soft bezel key on the FD MODE SEL panel.
 - a. Confirm APP P-ILS is displayed on the FD MODE SEL panel.

NOTE

LOC must be armed or captured before selecting GS. GS must be captured from below the GS. GS must be armed or captured before selecting DCL. Adjusting airspeed after activating DCL will cancel that function and requires rearming the DCL mode.

13. Select the LOC, GS and DCL soft bezel keys.
 - a. Confirm ARM appears in white for each mode (LOC, GS and DCL) on the FD MODE SEL panel.
 - b. Confirm that PLOC and PG/S are displayed in white under the P, R and C cues on the pilot PFD.
14. The helicopter will capture and continue to fly the flight plan to include the initial portion of the basic FMS approach. Once the helicopter intercepts the P-ILS inbound course, the NAV mode will automatically switch from FMS to P-ILS and the helicopter will fly the P-ILS approach. Confirm the following:
 - a. Smooth capture of the localizer with no more than a ½ dot overshoot.



- b. FD MODE SEL panel LOC turns from white ARM to green CAP.
- c. NAV light on FD MODE SEL panel extinguishes.
- d. LNAV on PFD is replaced with yellow P-ILS.
- e. The course deviation indicator on the HSI and the localizer indicator on the attitude indicator agree.
- f. Smooth capture of the glideslope with no more than a ½ dot overshoot.
- g. FD MODE SEL panel P-G/S turns from white ARM to green CAP.
- h. ALT light on FD MODE SEL panel extinguishes.
- i. ALT on PFD is replaced with green P-G/S.
- j. "MINIMUMS" aural alert is heard concurrently with MIN displayed in yellow on the PFD at the selected altitude ("EGPWS" aural altitude alert).
- k. "ALTITUDE" aural alert is heard at the selected RA altitude ("EGPWS" aural altitude alert).

NOTE

Auto level off and deceleration will not function unless the radar altimeter is on and providing a valid signal.

- l. Helicopter levels off at 50 feet AGL while maintaining localizer and 60 KIAS. DCL is replaced with IAS on the PFD and MSP.
15. Select Go Around mode by pressing the GA button on the pilot's cyclic. Confirm the following:
- a. Helicopter transitions to an 80 KIAS, 750 fpm climb on the last heading used to maintain localizer.
 - b. GA ON is displayed in green on both FD MODE SEL panels.
 - c. The pilot may then select HDG, IAS and VS or ALT at the desired values during GA.
16. Decouple by pressing the FD STBY button on the copilot cyclic.
- a. Confirm the "DECOUPLE" aural alert is heard.



4.6 ENHANCED GROUND PROXIMITY WARNING SYSTEM

4.6.1 ALTITUDE AURAL WARNINGS

NOTE

The "ONE HUNDRED" aural warning is a hard altitude warning.

1. Using the DCP, set MIN to 500 feet above landing zone.
2. Set RA to 250 feet AGL.
3. Conduct an approach to landing from greater than 650 feet AGL.
 - a. Verify that the aural warning "MINIMUMS" is heard on the ICS as the helicopter passes through the preset MIN $\pm 10\%$.
 - b. Verify that the aural warning "ALTITUDE" is heard on the ICS as the helicopter passes through the preset RA $\pm 10\%$.
 - c. Verify that the aural warning "ONE HUNDRED" is heard on the ICS as the helicopter passes through 100 ± 10 feet.

4.6.2 LANDING GEAR WARNINGS

NOTE

The helicopter needs to initially be above 400 AGL to activate the aural alert.

1. With the gear up and at an altitude above 150 feet AGL, slow below 60 KIAS.
 - a. Verify that no gear warning is heard.
2. Descend below 150 feet.
 - a. Verify that the red LNDG GEAR lights on both master warning panels illuminate.
 - b. Verify that the aural "TOO LOW GEAR" alert is heard.
 - c. Verify that each PFD displays GEAR.
3. Lower the landing gear.
 - a. Verify that the red UNLKD light on the landing gear panel illuminates while the landing gear is in transition.



- b. Verify that the three green lights on the landing gear panel illuminate.
- c. Verify that the red LNDG GEAR lights on the master warning panels extinguish.

4.6.3 TERRAIN DISPLAY

NOTE

Refer to the EGPWS pilot manual for appropriate display color band and density definitions.

1. Select NAV display.
2. Select OVLY.
3. Select TER.
4. Select a display range of 2.5 to 25 nm on the DCP.
 - a. Verify that the terrain is displayed.
 - b. Verify that the terrain display pattern colors and densities change as the helicopter climbs or descends.

4.6.4 OBSTACLE WARNING

NOTE

GPS look ahead modes are disabled under 70 KIAS.

NOTE

The "WARNING OBSTACLE" aural alert will activate when the helicopter is within 20 seconds of flight time from the obstacle.

1. Configure cockpit MFDs so that no EGPWS overlay is displayed anywhere in the cockpit.
2. Select a local target obstacle such as a radio tower.
3. Position the helicopter 4-5 nm from the target obstacle.
4. Fly toward the obstacle at a ground speed greater than 70 KIAS and at an altitude lower than the target obstacle.
5. At approximately 30 seconds of flight time from the target obstacle:
 - a. Verify that an MFD automatically displays a terrain display on a NAV page.
 - b. Verify that the terrain display range is automatically set to 5 nm.



- c. Verify that the amber OBSTACLE is displayed on the PFD.
- d. Verify that the aural warning "CAUTION OBSTACLE" is heard.
- e. Verify that approximately 7 seconds after the initial aural alert, a second "CAUTION OBSTACLE" aural warning is heard.

4.7 VH CHECK

NOTE

In certain high hot conditions, a TGT limit of 935°C may be reached before the 86% torque limit.

NOTE

In certain very cold conditions, a corrected Ng or fuel flow limit may be reached before the 86% torque limit.

1. Fly the helicopter straight and level and set engine torque 86%.
 - a. Verify that Nr is set to 105% Nr.
 - b. Verify that the hydraulic, main gear box and engine instruments on the EICAS page are in the green.
 - c. Verify that the aircraft exhibits no unusual vibration.
 - d. Verify that #1 and #2 torques are matched within 5%.
2. Select IAS and ALT on an AFCS MODE SEL panel.
3. Using the cyclic trim switch, increase desired airspeed to 155 KIAS.
 - a. Verify that engine torque is limited to 86% before the selected airspeed is achieved.
4. Deselect IAS and ALT and hand fly the helicopter.
5. Adjust power to maintain Vh.
 - a. Verify at least two inches of forward cyclic remain between the cyclic and the instrument panel.



5.0 EQUIPMENT CHECKS

5.1 TCAS

5.1.1 GROUND

1. Select NAV on an inboard MFD.
2. Select TCAS overlay.
3. Set the display range ring to 5nm using the DCP.
4. Select the HLTH→AVIONICS→SUB STAT→TCAS on another MFD.
 - a. Verify that the status is OPERATIONAL.
5. Press the IBIT bezel key on the left side of the MFD.
 - a. Verify that the status changes to TEST IN PROGRESS.
 - b. Verify that the TCAS-TEST is indicated on the MFD.
 - c. Verify that these 3 targets appear on the NAV display:
 - #1 – Traffic Advisory (yellow circle) will appear at 9 o'clock, range 2 miles, 200 feet below and climbing.
 - #2 – Proximity Traffic (solid cyan diamond) will appear at 1 o'clock, range 3.6 miles, 1000 feet below and descending.
 - #3 – Non-Threat Traffic (open cyan diamond) will appear at 11 o'clock, range 3.6 miles, flying level 1000 feet above.
 - d. Verify that the "TCAS SYSTEM TEST OK" aural alert is heard on the ICS.
 - e. Verify that the status changes to OPERATIONAL at the completion of the test.

5.1.2 FLIGHT

NOTE

Display symbology is delineated in the Flight Manual.



NOTE

Should TCAS 1 predict that certain safe boundaries might be violated, it will issue a Traffic Advisory as an aural alert and a display alert.

NOTE

Depending upon TCAS I sensitivity level, TCAS I will issue a Traffic Advisory when time to closest point of approach is 15 to 30 seconds.

1. At altitude, select NAV on an MFD.
2. Select OVLY.
3. Select TCAS.
4. Select CTL to activate TCAS controls.
5. Select BAND then NORM.
 - a. Verify that TCAS INOP is not displayed on the MFD.
6. Using targets of opportunity, monitor aircraft traffic.
 - a. Verify proper symbology for intruder aircraft.
 - b. If possible, verify that intruder aircraft are not ghosted.
 - c. If possible, verify that no phantom aircraft are displayed.
 - d. If possible, verify that the aural alert "TRAFFIC TRAFFIC" is heard when an intruder aircraft is near and closing.
 - e. If possible, verify that the MFD displays TRAFFIC when an intruder aircraft is near and closing.

5.2 FLOTATION

WARNING

To avoid possible injury in the event of inadvertent float deployment, clear the area around the nose, sponson (if installed) and tail floats prior to this test. All steps must be followed in sequence.



5.2.1 ORIGINAL FLOTATION CONTROL PANEL

NOTE

This procedure is applicable to the original flotation control panel in accordance with the Rotorcraft Flight Manual, Part 1.

CAUTION

To avoid possible injury in the event of inadvertent deployment, clear the area around the nose and tail floats prior to this test.

1. Primary FLOTATION toggle switch – TEST.
 - a. Confirm float ARMED (amber) and TEST (green) lights **not** illuminated.
2. FLOTATION TEST rotary switch – PRI.
 - a. Confirm float TEST (green) light illuminated.

CAUTION

Continue test only if the green TEST light is illuminated.

3. AUX FLOTATION toggle switch – ARM.
 - a. Confirm float ARMED (amber) light **not** illuminated.
 - b. Confirm float TEST (green) light remains illuminated.

CAUTION

Continue test only if the green TEST light is illuminated and neither L/R indicators are illuminated. Illumination of any L/R indicators may be an indication of a faulty immersion switch. With the ARMED (amber) light illuminated, a faulty immersion switch may cause immediate float deployment.

4. Pilot collective EMER FLOATS – Push.
 - a. Confirm the two L and two R (green) lights illuminated.



5. Copilot collective EMER FLOATS – Push.
 - a. Confirm the two L and two R (green) lights illuminated.
6. FLOTATION TEST rotary switch – SEC.
 - a. Confirm float ARMED (amber) light remains **not** illuminated.
 - b. Confirm the TEST (green) light remains illuminated.
7. Pilot collective EMER FLOATS – Push.
 - a. Confirm the two L and two R (green) lights illuminated.
8. Copilot collective EMER FLOATS – Push.
 - a. Confirm the two L and two R (green) lights illuminated.
9. FLOTATION TEST rotary switch – R IMMRS.
 - a. Confirm float ARMED (amber) light remains **not** illuminated.
 - b. Confirm the TEST (green) light remains illuminated.
 - c. Confirm the two L and two R (green) lights illuminated.
10. FLOTATION TEST rotary switch – L IMMRS.
 - a. Confirm float ARMED (amber) light remains **not** illuminated.
 - b. Confirm the TEST (green) light remains illuminated.
 - c. Confirm the two L and two R (green) lights illuminated.
11. FLOTATION TEST rotary switch – NORM.
 - a. Confirm all TEST (green) lights extinguished.
12. FLOTATION toggle switch – SAFE.
 - a. Confirm float ARMED (amber) light illuminated.
13. AUX FLOTATION toggle switch – SAFE.
 - a. Confirm float ARMED (amber) light extinguished.
14. Primary FLOTATION toggle switch – ARM.
 - a. Confirm float ARMED (amber) light illuminated.
15. Primary FLOTATION toggle switch – SAFE.



- a. Confirm float ARMED (amber) light extinguished.

5.2.2 IMPROVED FLOTATION CONTROL PANEL

NOTE

This procedure is applicable to the improved flotation control panel in accordance with the Rotorcraft Flight Manual Supplement No. 1, Revision 2, Part 1.

CAUTION

To avoid possible injury in the event of inadvertent float deployment, clear the area around the nose, sponson (if installed) and tail floats prior to this test. All steps must be followed in sequence.

1. FLOTATION ARMING switch (center console) – SAFE.
2. FLOTATION TEST rotary switch (pilot overhead console) – OFF.
3. FLOTATION TEST switch (pilot overhead console) – TEST.

NOTE

Three sets of L and R lights on the flotation test panel are designed to identify the forward, aft and sponson float bags. When the sponson float bags are not installed, the middle L and R lights will not illuminate during the following test.

4. FLOTATION TEST rotary switch – PRI.
 - a. Confirm both float ARMED (amber) and EICAS FLOAT ARMED advisory or caution extinguished.
 - b. Confirm float TEST (green) light illuminated.

WARNING

Continue test only if the green TEST light is illuminated.

5. Pilot collective EMER FLOATS – Push.
 - a. Confirm the three L and three R (green) lights illuminated.



6. Copilot collective EMER FLOATS – Push.
 - a. Confirm the three L and three R (green) lights illuminated.
7. FLOTATION TEST rotary switch – R IMMRS.
 - a. Confirm the three L and three R (green) lights illuminated.
8. FLOTATION TEST rotary switch – SEC.
 - a. Confirm float ARMED (amber) light remains extinguished.
 - b. Confirm EICAS FLOAT ARMED advisory or caution remains extinguished.
 - c. Confirm float TEST (green) light remains illuminated.
9. Pilot collective EMER FLOATS – Push.
 - a. Confirm the three L and three R (green) lights illuminated.
10. Copilot collective EMER FLOATS – Push.
 - a. Confirm the three L and three R (green) lights illuminated.
11. FLOTATION TEST rotary switch – L IMMRS.
 - a. Confirm the three L and three R (green) lights illuminated.
12. FLOTATION TEST rotary switch – OFF.
13. FLOTATION TEST switch – NORMAL.
 - a. Confirm both float ARMED (amber) lights extinguished.
 - b. Confirm EICAS FLOAT ARMED advisory or caution remains extinguished.
14. FLOTATION ARMING switch – ARM.
 - a. Confirm both float ARMED (amber) lights illuminated.
 - b. Confirm EICAS FLOAT ARMED caution illuminated.
15. Primary FLOTATION toggle switch – SAFE.
 - a. Confirm both float ARMED (amber) lights extinguished.
 - b. Confirm EICAS FLOAT ARMED advisory or caution extinguished.



5.3 LIFE RAFT

WARNING

To avoid possible injury in the event of inadvertent deployment, clear the area around the left hand and right hand forward sponson life raft compartment prior to these tests.

5.3.1 ORIGINAL LIFE RAFT CONTROL PANEL

NOTE

This procedure is applicable to the original life raft control panel in accordance with the Rotorcraft Flight Manual, Part 2.

1. LIFE RAFT TEST switch – PRI.
 - a. Confirm raft ARMED (amber) light **not** illuminated.
 - b. Confirm raft TEST (green) light **not** illuminated.
2. LIFE RAFT switch – ARM.
 - a. Confirm raft ARMED (amber) light illuminated.
 - b. Confirm raft TEST (green) light illuminated.

WARNING

Continue test only if the green TEST light is illuminated.

3. JETT switch – Actuate.
 - a. Confirm the L and R (green) lights illuminated.
4. JETT switch – Release.
 - a. Confirm the L and R (green) lights extinguish.
5. LIFE RAFT TEST switch – SEC.
 - a. Confirm raft ARMED (amber) light illuminated.



- b. Confirm raft TEST (green) light illuminated.
- 6. JETT switch – Actuate.
 - a. Confirm the L and R (green) lights illuminated.
- 7. JETT switch – Release.
 - a. Confirm the L and R (green) lights extinguish.
- 8. LIFE RAFT TEST switch – NORM.
 - a. Confirm raft ARMED (amber) light illuminated.
 - b. Confirm raft TEST (green) light **not** illuminated.
- 9. LIFE RAFT switch – SAFE.
 - a. Confirm raft ARMED (amber) light **not** illuminated.
 - b. Confirm raft TEST (green) light **not** illuminated.

5.3.2 IMPROVED LIFE RAFT CONTROL PANEL

NOTE

This procedure is applicable to the improved life raft control panel in accordance with the Rotorcraft Flight Manual, Supplement No. 1, Part 2.

- 1. LIFE RAFT toggle switch – TEST.
- 2. Rotary TEST switch – PRI.
 - a. Confirm raft TEST (green) light illuminated.
 - b. Confirm raft green ARM light and MFD RAFT ARM cautions are extinguished.

WARNING

Do not continue if either ARM light or RAFT ARM caution is illuminated or the TEST light is not illuminated.

- 3. JETT switch – Actuate.
 - a. Confirm the L and R (green) lights illuminated.



4. JETT switch – Release.
 - a. Confirm the L and R (green) lights extinguish.
5. Rotary TEST switch – SEC.
 - a. Confirm raft TEST (green) light illuminated.
 - b. Confirm raft green ARM light and MFD RAFT ARM cautions are extinguished.
6. JETT switch – Actuate.
 - a. Confirm the L and R (green) lights illuminated.
7. JETT switch – Release.
 - a. Confirm the L and R (green) lights extinguish.
8. Rotary TEST switch – OFF.
9. LIFE RAFT toggle switch – SAFE.
 - a. Confirm raft TEST light extinguished.
 - b. Confirm both ARM light and the MFD RAFT ARM cautions are extinguished.
10. LIFE RAFT toggle switch – ARM.
 - a. Confirm both ARM light and the MFD RAFT ARM cautions are illuminated.
11. LIFE RAFT toggle switch – SAFE.
 - a. Confirm both ARM light and the MFD RAFT ARM cautions are extinguished.

5.4 RESCUE HOIST

NOTE

These procedures are applicable to single or dual rescue hoist installation, and single or dual pendant installation.

NOTE

If connected both pendants should be active at the same time and operate the active hoist.



5.4.1 HOIST SYSTEM SHEAR TEST

1. REQ HST switch – ALL.

NOTE

The yellow CAUTION light on the pendent will illuminate whenever the selected hoist hook is within 15.5 feet from the hoist motor (near fully retracted) or when only 7 – 7.5 wraps of cable remain on the storage drum (near fully extended).

2. HOIST ACTIVE – INBOARD.
3. SHEAR TEST switch – PRI.
 - a. Confirm SHEAR STATUS – TEST.



To prevent cable shear, do not activate any SHEAR control unless the SHEAR TEST switch is in either the PRI or SEC position and the green SHEAR STATUS – TEST light is illuminated.

NOTE

The SHEAR STATUS – ARMED light will remain illuminated during the shear test.

4. Rescue hoist panel HOIST SHEAR switch – Activate.
 - a. Confirm SHEAR TEST – PRI light illuminates.
5. Pilot cyclic HOIST SHEAR switch – Activate.
 - a. Confirm SHEAR TEST – PRI light illuminates.
6. Copilot cyclic HOIST SHEAR switch – Activate.
 - a. Confirm SHEAR TEST – PRI light illuminates.
7. SHEAR TEST switch – SEC.
 - a. Confirm SHEAR STATUS – TEST.
8. Rescue hoist panel HOIST SHEAR switch – Activate.
 - a. Confirm SHEAR TEST – SEC light illuminates.



9. Pilot cyclic HOIST SHEAR switch – Activate.
 - a. Confirm SHEAR TEST – SEC light illuminates.
10. Copilot cyclic HOIST SHEAR switch – Activate.
 - a. Confirm SHEAR TEST – SEC light illuminates.
11. SHEAR TEST switch – NORM.
 - a. Confirm SHEAR STATUS – ARMED.
 - b. Confirm SHEAR TEST light **not** illuminated.
12. For dual hoist installation: HOIST ACTIVE – OUTBOARD.
13. For dual hoist installation: Repeat steps 3. – 11. above.
14. RSQ HST – OFF.

5.4.2 HOIST SYSTEM OPERATIONAL TEST

1. HOIST ACTIVE – INBOARD.
2. Confirm that the hoist shear test has been successfully completed by either maintenance personnel or pilots.
3. Place the RSQ HST switch in the ALL position.
 - a. Verify that the HOIST ARMED advisory light illuminates.
 - b. Verify that the ARMED light illuminates on the cabin rescue hoist panel.
4. Extend and retract the cable using the hoist operator's control panel.
5. Extend the cable using the rescue hoist pendant.
6. While crewman is extending, push the RSQ HOIST switch on the pilot's cyclic UP.
 - a. Verify that the cable retracts, overriding the crewmember's DOWN command.
7. While crewman is extending, push the RSQ HOIST switch on the copilot's cyclic UP.
 - a. Verify that the cable retracts, overriding the crewmember's DOWN command.
8. Have a crewmember fully retract the cable.
9. For dual pendant installation, extend the cable using the other rescue hoist pendant.
10. For dual pendant installation, repeat steps 6. – 8.



11. Place the RSQ HST switch in the OFF position.
 - a. Verify that the HOIST ARMED advisory light extinguishes.
 - b. Verify that the ARMED light extinguishes.
12. For dual hoist installation: HOIST ACTIVE – OUTBOARD.
13. For dual hoist installation: repeat steps 2. – 11. above.

5.5 CARGO HOOK

NOTE

There are three manual cargo hook releases. One release is on top of the hook (accessible from the cabin), the second is on the side of the hook (accessible from under the helicopter) and the third is the copilot's manual CARGO RELEASE.

1. Place the CARGO HK switch in the ARM position.
 - a. Verify that the HOOK ARMED advisory light illuminates.
2. Place approximately 20 lbs download pressure on the hook.
3. Press and release the CRG REL button on the pilot's cyclic.
 - a. Verify that the HOOK OPEN advisory light momentarily illuminates.
 - b. Verify that the cargo hook opens.
4. Repeat for the copilot's cyclic release.
5. Place the CARGO HK switch in the SAFE position.
 - a. Verify that the HOOK ARMED advisory light extinguishes.
6. Place approximately 20 lbs download pressure on the hook.
7. Pull the cockpit CARGO HOOK RELEASE handle.
 - a. Verify that the HOOK OPEN advisory light momentarily illuminates.
 - b. Verify that the cargo hook opens.
8. Repeat using the manual cargo release lever located on the cargo hook assembly.



5.6 RIPS

NOTE

The Rotor Ice Protection System (RIPS) IBIT may be conducted with RIPS in OFF, AUTO or MANUAL mode.

NOTE

The RIPS IBIT heats the blades for a short period of time. There is no OAT limit for the RIPS IBIT. However, a lockout function inhibits a subsequent IBIT if requested within approximately 90 seconds of the last IBIT. The system status will display TEST INHIBIT if this lockout function is activated.

NOTE

A RIPS system RESET may be initiated before a subsequent IBIT if the RIPS IBIT fails. RESET removes all power from the system (like pulling CB) and initiates PBIT. The PBIT takes approximately 90 seconds. An IBIT cannot be conducted until the PBIT is completed.

NOTE

Both engines must be in FLY to successfully complete an IBIT.

NOTE

Engine inlets with ice detectors must be installed to conduct a RIPS IBIT.

1. Set 105% Nr at flat pitch with both engines in FLY.
2. Select both main generators ON.
3. Select the HLTH page.
4. Select RIPS.
5. Select IBIT.
 - a. Note a momentary display of the ice rate meter on the MFD.
 - b. Verify that the RIPS system status is TEST IN PROGRESS.
 - c. Verify that the TEST indication illuminates under ICE on MFD.
6. When the IBIT is complete (approximately 1 min 15 sec):
 - a. Verify that the TEST indication under ICE on MFD changes to OFF, AUTO, or MANUAL depending on the mode selected when the test was initiated.



- b. Verify that the RIPS status changes to TEST PASSED.
- c. Verify that the RIPS system status changes to OPERATIONAL.

5.7 WEATHER RADAR



The weather radar has a weight on wheels override that forces the system into standby when the aircraft is on the ground. Unless STBY or OFF are selected, the weather radar will radiate as soon as the helicopter lifts to a hover. If an active radar mode is selected, ensure all people are at least 100 feet away from the forward 270° arc of the aircraft prior to takeoff.

NOTE

These weather radar procedures are to be conducted when the helicopter is airborne. Range displayed on inner range ring is one-half range.

NOTE

A time delay prevents the transmitter from operating until the magnetron has warmed up. The radar displays WAIT and does not transmit or scan the antenna for the approximate 90-second warm-up period. It then automatically becomes operational in the selected mode or goes to FSBY if the helicopter is on the ground.

NOTE

In the TST mode, the transmitter output power radiates only if FSBY is defeated. Observe the display and verify that TEST is displayed in the WxR MODE field and the range and alphanumeric marks are displayed in white.

1. Select NAV on one MFD.
2. Select OVLY and WXR from soft bezel keys.
3. Set the weather radar mode switch to TST.
 - a. Verify test pattern appears and radar begins scanning. Also note alternate scanning of the pilot side vs. copilot side ARC displays (this verifies proper configuration strapping).
4. Set the weather radar mode switch to WX.



5. Push the radar control panel GAIN knob in to select automatic gain.
6. Adjust TILT on the weather radar control panel and RANGE on the DCP.
 - a. Verify WX is displayed on the MFD.
7. Pull the weather radar control panel GAIN knob to select manual gain.
8. Rotate the radar control panel GAIN knob.
 - a. Verify that gain value adjusts on the display.
9. Push the weather radar control panel GAIN control knob in to re-select automatic gain.
10. Rotate the radar control panel TILT control knob from full clockwise to full counterclockwise.
 - a. Verify that the digital TILT annunciation on the display changes smoothly from 15 at the full clockwise position to -15 at the full counterclockwise position.
 - b. Verify the antenna transitions from approximately +15 to -15° by observing the display.
11. Select different ranges (2.5, 5, 12.5, 25, 50, 100, 150) using the RANGE knob on the DCP.
 - a. Verify that the ranges change on the MFD.
12. On the radar control panel, push the SECT button.
 - a. Verify the displayed scan is reduced to a 60° sector with a scan rate of approximately one sweep every 2.5 seconds.
13. Push the radar control panel SECT button again.
 - a. Verify that the scan is restored to the 120° scan rate of approximately one sweep every 5 seconds.
14. Push the radar control panel TGT button.
 - a. Verify that TGT (Target Alert Mode) appears on the MFD.
15. Push the radar control panel TGT button again.
 - a. Verify that the TGT disappears.

5.8 LIGHTNING STRIKE SENSOR

NOTE

This procedure is for the Honeywell LSZ-860 displayed on a Collins MFD.



NOTE

The IBIT performs a full verification of the system by generating a known signal in the antenna and displaying the results on the weather radar display.

NOTE

The IBIT can be conducted with either weather or terrain overlaid on the NAV display.

1. Select NAV and FULL on one MFD.
2. Set the weather radar mode switch to WX.
3. Select 50 NM or greater range.
4. Select OVLY and WX or TER.
5. Select LTNG.
6. Select HLTH on another MFD.
7. Select AVIONICS, SUB STAT and LSS.
 - a. Verify that the Lightning Strike Sensor (LSS) status is OPERATIONAL.
8. Select the IBIT bezel key.
 - a. Verify that the LSS status changes to TEST IN PROGRESS.
 - b. Verify that any active lightning rate symbols are erased from the WX display.
 - c. Verify that the LSS mode is in LTNG TEST.

NOTE

A simulated lightning test pulse is sent to the antenna after 3 to 4 seconds. It then takes 5 to 7 seconds to build the rate 3 symbol on the display.

- d. Verify that a rate 3 (white) lightning symbol is displayed at 25 ± 5 NM and at 45° to the right of the nose.
- e. Verify that a rate 1 (magenta) lightning alert symbol is displayed and remains on for 3 to 7 seconds at the maximum selected range and at 45° to the right of the nose.
- f. Verify that the LSS status changes to TEST PASSED.
- g. Verify that the LSS status returns to OPERATIONAL.



5.9 DIGMAP

NOTE

The DigMap consists of 2 independent systems; one on the pilot's side and the other on the copilot's side.

NOTE

DigMap can be displayed on any of the MFDs.

NOTE

DigMap is not FAA certified for navigation.

NOTE

The display of the range rings confirms all the DigMap interfaces (FMS and AHRS) are inputted properly.

NOTE

A Clear BIT Fail Flag must be initiated to erase any stored faults before a BIT check is run.

1. On the pilot's inboard MFD, select VIDEO.
2. Select DigMap.
 - a. Verify that the display is clear and consistent with the range selected.
3. Using the RANGE control knob on the pilot's DCP, verify that the range can be changed.
4. Using the pilot's DigMap control panel, select MEN.
5. Using the arrow cursor, select Maintenance→ENT.
6. Select BIT→ENT.
7. Select Clear BIT Fail Flag→ENT→ENT (Yes).
8. Select Run BIT→ENT. The BIT takes approximately 15 seconds to complete.
9. When the BIT is complete, select View BIT Log→ENT.
10. If no file is logged, the BIT has passed.
11. If a file is logged, note the fault.
12. Select ESC to exit menu.



13. Repeat for the copilot's side.

5.10 SAR AFCS

5.10.1 SAR HOVER CHECKS

1. Select YAW, CYCLIC, and COLL TRIM ON.
2. Select ATT, AP2, SAS2, and PRI ON.
3. Select SAS1 and PRI ON.
4. Select AP1 OFF.
5. Select FD2 on the AFCS control panel.
6. Select a target hover altitude by adjusting HVR RA on the pilot's DCP to 50 feet.
 - a. Verify that the radar altitude bug appears on the radar altimeter display.
7. Establish stable hover with minimal drift.
8. Select HOV on the PFD.
9. Select SAR→VHLD→PHLD on an FD MODE SEL panel.
 - a. Verify that FD2, P HOV, R HOV, C RALT and S PHLD display on the PFD header.
 - b. Verify that drift velocities are displayed on the PFD below the hover display.
 - c. Verify that the aircraft maintains 50 feet.
 - d. Verify that the aircraft hovers over the selected position.
10. Without depressing any trim, displace cyclic to drag the aircraft at least 30 feet away from the target hover position.
 - a. Verify that the target star and range (RNG) display correctly on the PFD.
11. Release cyclic.
 - a. Verify that the aircraft smoothly returns to the target hover position.
12. Initiate a 5 knot forward drift using the pilot cyclic beeper trim.
 - a. Verify that VHLD replaces PHLD on the PFD header.



- b. Verify that the velocity vector and the acceleration ball on the hover display respond correctly.
13. Push down on the pilot cyclic beeper.
 - a. Verify that PHLD re-engages.
 - b. Verify that the aircraft slows to less than 2 knots and then the AHRS determines a new target hover position.
 - c. Verify that the aircraft hovers at the new target position.
14. Repeat initiate 5 knot drift in aft, left and right directions.
15. From a coupled hover, depress collective trim and descend to below 30 feet.
 - a. Verify that the "HOVER ALTITUDE" aural alert is heard continuously by both pilots.
16. Return to hover altitude.
17. From a coupled hover at a selected altitude, without depressing the collective trim, descend 5 feet and slowly release collective.
 - a. Verify that the helicopter returns to selective altitude within ± 4 feet with no more than a 4 foot overshoot.
18. Select another altitude by adjusting the collective beeper.
 - a. Verify that the aircraft climbs/descends to the new altitude.
19. Select ALT on the FD MODE SEL panel.
 - a. Verify that ALT (barometric altimeter) replaces RALT on the PFD header.
 - b. Verify that the aircraft maintains hover altitude.
20. Repeat the entire procedure using AP1, FD1 and the copilot cyclic/collective beepers.

5.10.2 SAR DEPARTURE

1. Select YAW, CYCLIC, and COLL TRIM ON.
2. Select ATT, AP2, SAS2, and PRI ON.
3. Select SAS1 and PRI ON.
4. Select AP1 OFF.
5. Select FD 2 on the AFCS control panel.
6. Establish a stable hover.



7. Push the DPT/GA button on the pilot's cyclic.
 - a. Verify that FD2, P DPT, R (blank), C DPT and S DPT display on the PFD header.
 - b. Verify that the aircraft climbs at approximately 200 fpm and slowly accelerates (2.5 kts/sec).
 - c. Verify that the aircraft transitions to GA at 65 KIAS.
8. Repeat with AP1, FD1, and copilot's cyclic.

5.10.3 SAR APPROACH TO POINT

NOTE

This check can be conducted over a long runway or over water. When using a runway, designate a target point at the end of the runway and initiate a straight-in ATPT over the approach end of the runway at a low enough altitude to arrive at the target point.

1. Select YAW, CYCLIC, and COLL TRIM ON.
2. Select ATT, AP2, SAS2, and PRI ON.
3. Select SAS1 and PRI ON.
4. Select AP1 OFF.
5. Select FD2 on the AFCS control panel.
6. Select NAV display.
7. Select bearing pointer information on the PFD (do not select the PFD analog radar altimeter).
8. Designate an approach to hover point using either the FMS or the middle button on the slew controller.
9. Select the desired hover altitude by adjusting HVR RA on the pilot display control panel.
 - a. Verify that the desired hover height appears on the PFD.
10. Position the aircraft for the approach (straight-in over a runway or anywhere over the water).

NOTE

If the aircraft is too high or fast to intercept the approach, IAS and RALT will engage and the aircraft will continue straight ahead with no ATPT symbology.

11. Set airspeed above 60 KIAS and altitude between 200 and 2000 feet AGL.



12. Select SAR→ATPT on the FD MODE SEL panel.

NOTE

IAS and RALT might initially engage until the aircraft decelerates and/or descends.

- a. Verify that FD2, P APP, R APP, C APP (RALT underneath), and S ATPT appears on the PFD header.
 - b. Verify that navigation information appears on the PFD and NAV page.
 - c. Verify that the analog radar altimeter automatically appears on the PFD.
 - d. Verify that the HOV display on the PDF automatically appears below 50 knot ground speed.
 - e. Verify that the aircraft intercepts the approach line and ends up in a hover directly over the target position at the selected altitude.
 - f. Verify that the aircraft transitions to PHLD over the target position.
13. Repeat with AP1 and FD1.

5.10.4 SAR MARK ON TOP

1. Select YAW, CYCLIC and COLL TRIM ON.
2. Select ATT, AP2, SAS2 and PRI ON.
3. Select SAS1 and PRI ON.
4. Select AP1 OFF.
5. Select FD2 on the AFCS control panel.
6. Select the desired hover altitude by adjusting HVR RA on the pilot DCP.
 - a. Verify that the desired hover height displays on the PFD.
7. Set airspeed above 80 KIAS and altitude between 300' and 2,000' AGL.
8. Position the aircraft to fly over the target.
9. When over target, select SAR→MOT on the FD MODE SEL panel.
 - a. Verify that FD2, P APP, R APP, C APP (RALT underneath) and S MOT display on the PFD header.
 - b. Verify that APP1 mode is activated on the FD MODE SEL panel.



- c. Verify that the previous FMS flight plan is removed from the NAV display and replaced with the MOT pattern, with the approach final approach path into the prevailing wind.
 - d. Confirm that the aircraft descends to 200' AGL, slows to 80 KIAS and follows the MOT pattern.
 - e. Verify that IAS mode is activated on the FD MODE SEL panel when aircraft reaches 80 KIAS.
 - f. Verify that RALT engages on the FD MODE SEL panel at 200' AGL.
 - g. Confirm that the aircraft intercepts the approach line.
 - h. Confirm that the aircraft begins to descend and slows after the final approach fix.
 - i. Verify that C APP (RALT underneath) label displays on the PFD header on the FD MODE SEL panel.
 - j. Verify that APP2 mode is activated on the FD MODE SEL panel.
 - k. Verify that the HOV display on the PFD automatically displays below 50 knot ground speed.
 - l. Confirm that the aircraft ends up in a hover approximately 50 yards to the left and 50 yards downwind of the target position at the selected RA altitude.
 - m. Verify that PHLD and VHLD are activated on the FD MODE SEL panel at the end of the approach.
 - n. Verify that P HOV, R HOV, C RALT and S PHLD display on the PFD header while in a coupled hover.
10. Repeat with AP1 and FD1.

5.10.5 SAR CREW HOVER

1. Select YAW, CYCLIC, and COLL TRIM ON.
2. Select ATT, AP2, SAS2, and PRI ON.
3. Select SAS1 and PRI ON.
4. Select AP1 OFF.
5. Select FD 2 on the AFCS control panel.
6. Establish stable hover above 30 feet with minimal drift.
7. Select HOV on the PFD.
8. Select SAR→VHLD→CHOV on a FD MODE SEL panel.
 - a. Verify that FD2, P HOV, R HOV, C RALT, and C CHOV appears on the PFD header.
 - b. Verify that the crew hover drift velocities are displayed on the PFD below the hover display.



- c. Verify that the HOVER light on the cabin hoist control panel illuminates.
9. Have crewman initiate a 10 knot forward drift using the crew hover cyclic beeper.
 - a. Verify that the velocity vector and the acceleration ball on the hover display respond correctly.
10. Push down on the pilot cyclic beeper for greater than 1 second.
 - a. Verify that VHLD engages and then PHLD engages.
 - b. Verify that the aircraft hovers at the new position.
11. Repeat initiate 10 knot drift in aft, left and right directions.
12. Repeat the entire procedure using AP1, FD1, and the copilot cyclic beeper.

5.11 RIG APPROACH

NOTE

This check can be conducted over water using a waypoint to simulate an offshore rig.

1. Select YAW, CYCLIC, and COLL TRIM ON.
2. Select ATT, AP2, SAS 2, and PRI ON.
3. Select SAS 1 and PRI ON.
4. Select AP1 OFF.
5. Select FD 2 on the AFCS control panel.
6. Select a suitable FMS waypoint (simulated rig) to be used as a rig location.
7. Set simulated rig waypoint as the next and final active waypoint.
8. Set LNAV as the navigational mode using the pilot's DCP.

NOTE

RIG preview can only be selected on the DCP corresponding to the active flight director.

9. Select RIG preview on the pilot's DCP.
10. Set inbound course with the Remote Instrument Controller (RIC).
11. Set the offset direction and distance on the Mode Select Panel (MSP) (MSP→APP→RIG).



12. Set the Minimum Descent Height (MDH) to 200 feet via HVR RA selection on the DCP.
13. Position aircraft above 2,500 feet AGL, 120 KIAS, and approximately 15 NM from the rig location and not on the final approach course.

NOTE

If the rig approach is activated without sufficient time to accomplish a normal descent and deceleration prior to the Turn Point (TP), the TOO HIGH or TOO CLOSE alert will display on the MSP.

14. Select APP on the MSP to access the rig approach page.
15. Activate the approach by selecting RIG on the pilot's MSP.
 - a. Verify LOC, GS, and DCL are ARMED on the DCP.
 - b. Verify that the aircraft initially maintains current airspeed and barometric altimeter.
 - c. Verify that the aircraft descends to arrive at the TP at 1,500 feet AWL with a maximum rate of descent of 750 fpm.
 - d. Verify that the aircraft decelerates to arrive at the TP at 80 KIAS.
16. At the TP:
 - a. Verify that the aircraft is at 1,500 feet AWL.
 - b. Verify that the aircraft is at 80 KIAS.
17. At the IAF:
 - a. Verify that the aircraft intercepts final approach course (localizer).
 - b. Verify that the aircraft is at 1,500 feet AWL.
 - c. Verify that the aircraft is at 80 KGS.
18. At FAF:
 - a. Verify that the aircraft descends on glideslope and localizer.
 - b. Verify that the aircraft decelerates to arrive at 30 KGS just prior to leveling at MDH.
19. At MAP:
 - a. Verify that the aircraft maintains localizer course.
 - b. Verify that the aircraft maintains 30 KGS in VHLD.
 - c. Verify that the aircraft is at MDH in RALT.



- d. Verify that the aircraft is at selected offset distance and direction.
20. Select GA.
- a. Verify that the aircraft maintains localizer course.
 - b. Verify that the aircraft transitions to 80 KIAS.
 - c. Verify that the aircraft transitions to 750 fpm climb.
21. Repeat with AP1, FD 1, copilot's DCP and copilot's MSP.



FIGURES

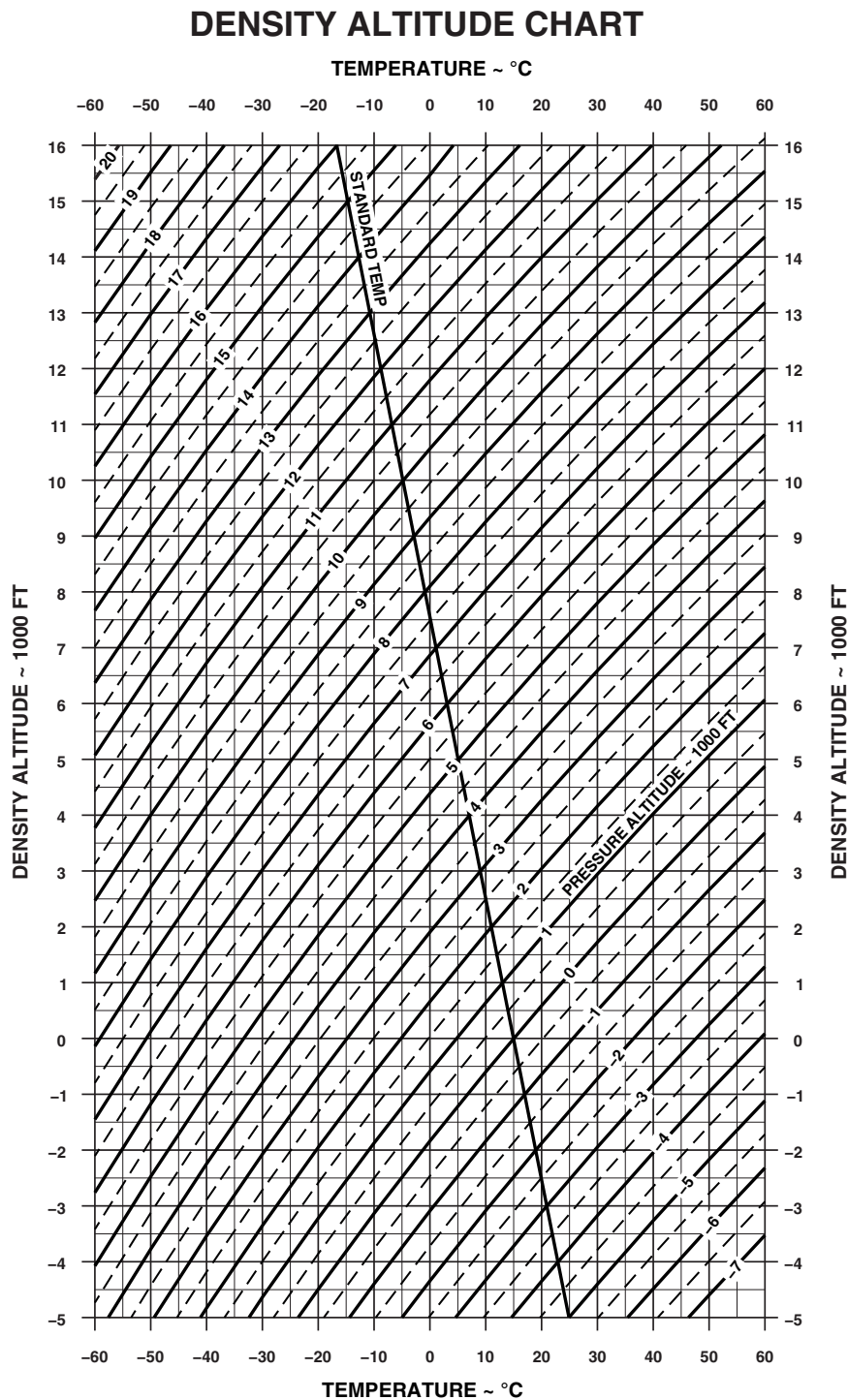
RF91110
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Figure 1. Density Altitude Chart

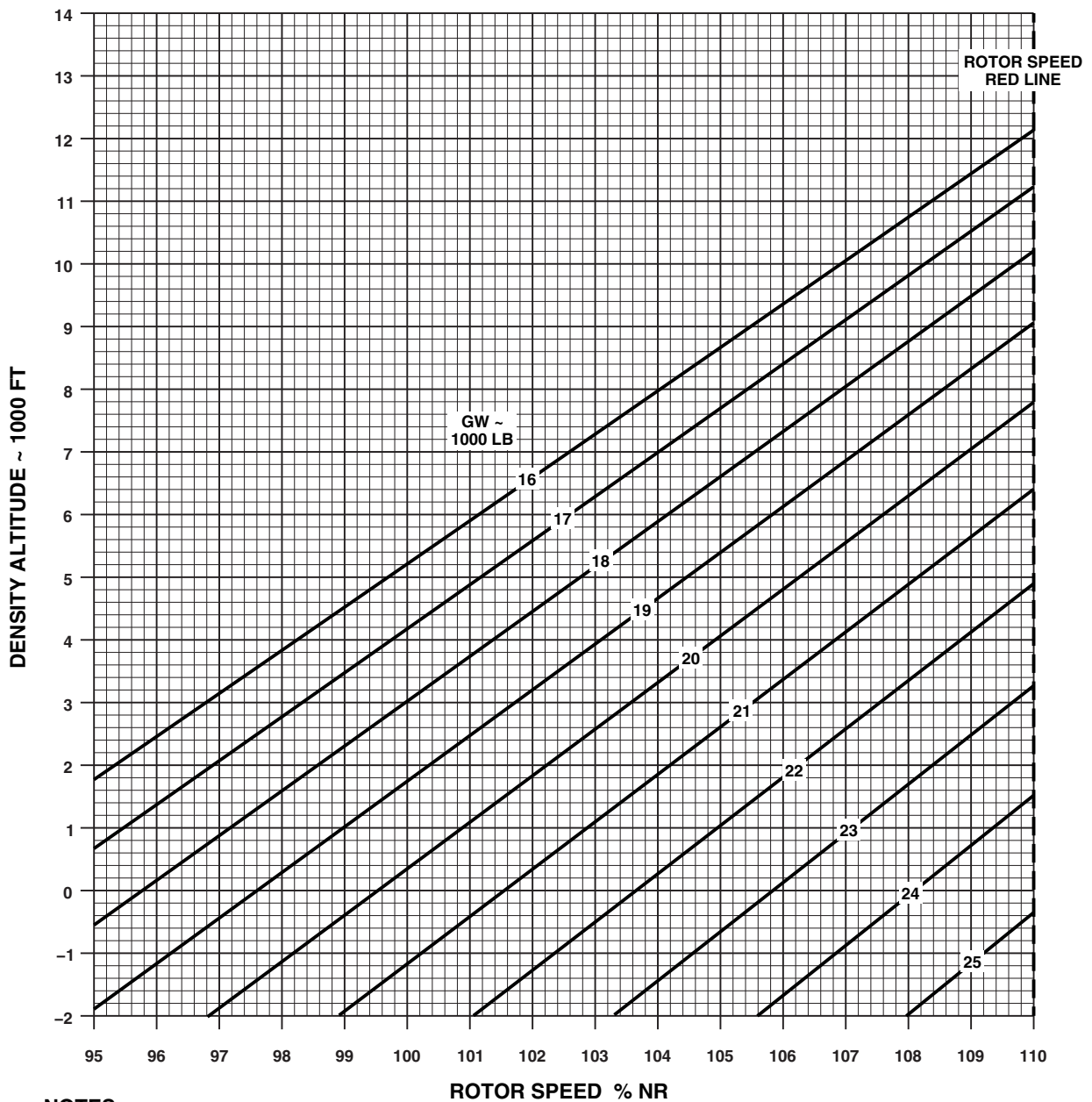
September 11, 2004

Revised: October 18, 2013



S-92 AUTOROTATION RPM CORRECTION CHART

10% COLLECTIVE



NOTES

1. OBSERVE NR ENVELOPE (95% NR - 110% NR)
2. COLLECTIVE TO 10%
3. TEST AIRSPEED 80 KIAS PILOT IAS (± 2 KTS)
4. AUTOROTATION TEST TOLERANCE IS $\pm 3\%$ NR
5. ROTOR SPEED DECREASES 4.5% PER FULL TURN OF ROTATING PUSHROD
6. ROTOR SPEED DECREASES 1% PER 8 INCHES AFT OF 352 IN
7. ROTOR SPEED INCREASES 1% PER 8 INCHES FORE OF 352 IN

RF91105_1B
SAF

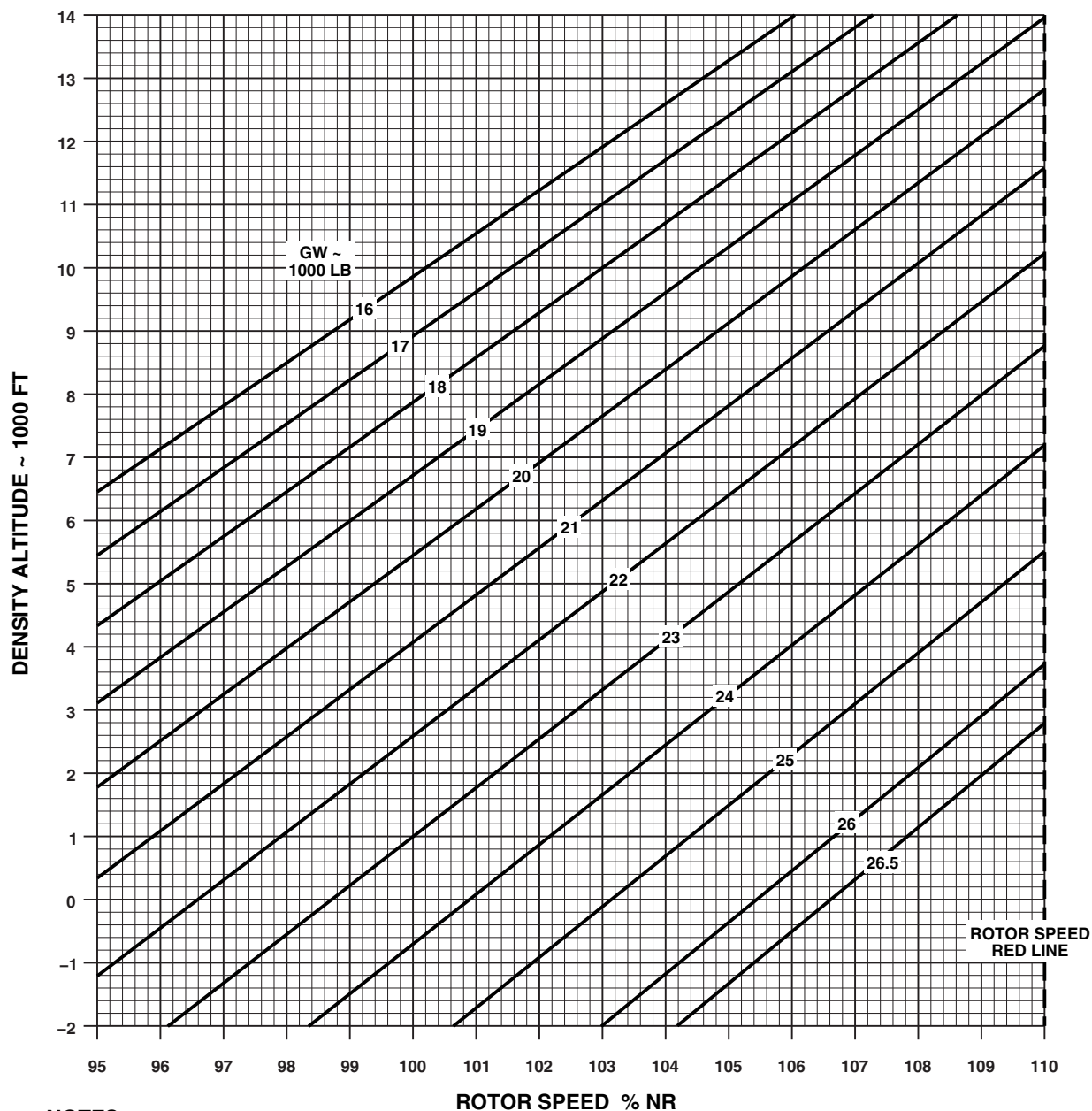
Figure 2. Autorotation RPM Correction Chart - 10%

September 11, 2004
Revised: October 18, 2013



S-92 AUTOROTATION RPM CORRECTION CHART

15% COLLECTIVE



NOTES

1. OBSERVE NR ENVELOPE (95% NR - 110% NR)
2. COLLECTIVE TO 15%
3. TEST AIRSPEED 80 KIAS PILOT IAS (± 2 KTS)
4. AUTOROTATION TEST TOLERANCE IS $\pm 3\%$ NR
5. ROTOR SPEED DECREASES 4.5% PER FULL TURN OF ROTATING PUSHROD
6. ROTOR SPEED DECREASES 1% PER 8 INCHES AFT OF 352 IN
7. ROTOR SPEED INCREASES 1% PER 8 INCHES FORE OF 352 IN

RF91105_2B
SAF

Figure 3. Autorotation RPM Correction Chart - 15%

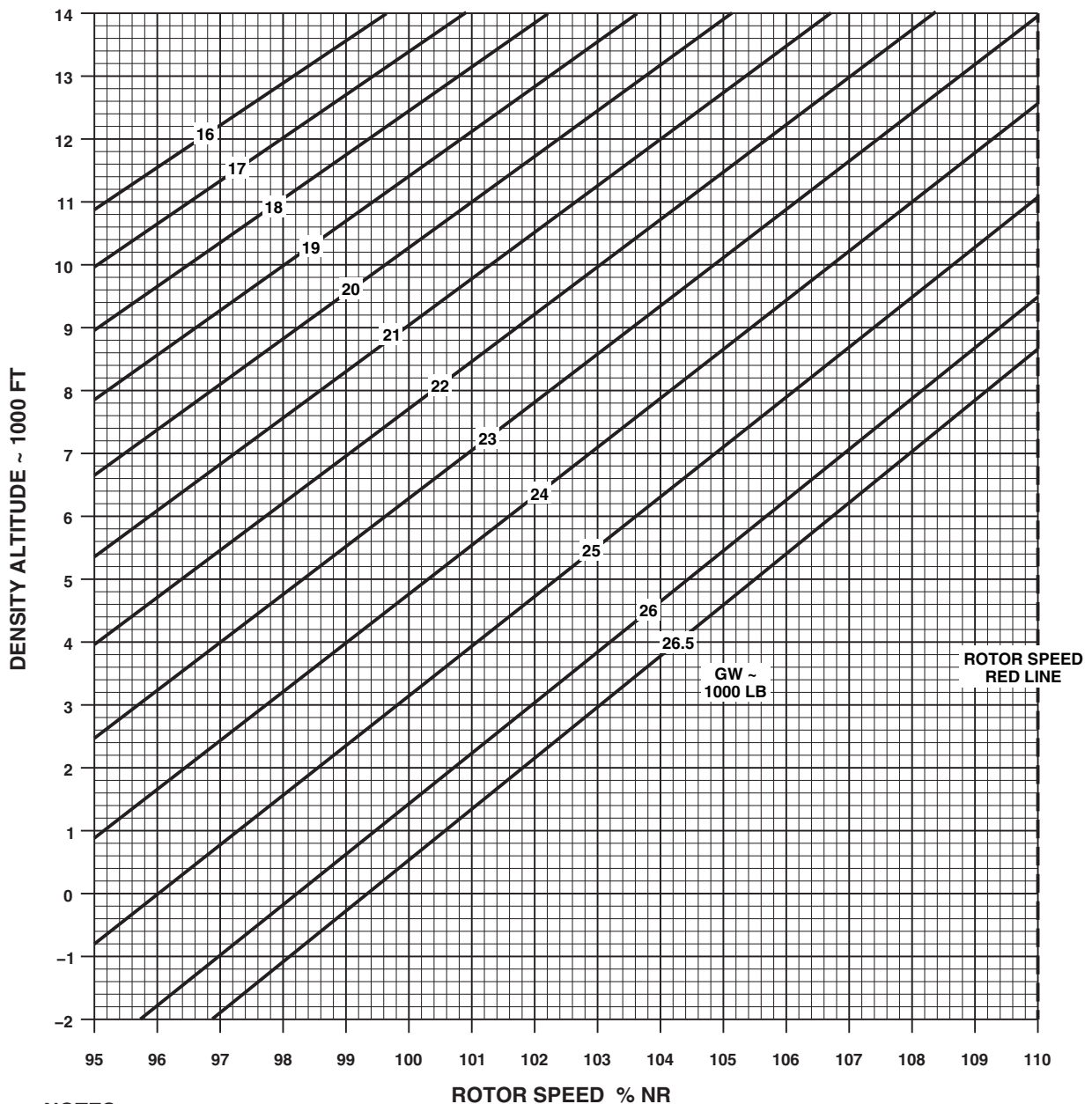
September 11, 2004

Revised: October 18, 2013



S-92 AUTOROTATION RPM CORRECTION CHART

20% COLLECTIVE



NOTES

1. OBSERVE NR ENVELOPE (95% NR - 110% NR)
2. COLLECTIVE TO 20%
3. TEST AIRSPEED 80 KIAS PILOT IAS (± 2 KTS)
4. AUTOROTATION TEST TOLERANCE IS $\pm 3\%$ NR
5. ROTOR SPEED DECREASES 4.5% PER FULL TURN OF ROTATING PUSHROD
6. ROTOR SPEED DECREASES 1% PER 8 INCHES AFT OF 352 IN
7. ROTOR SPEED INCREASES 1% PER 8 INCHES FORE OF 352 IN

RF91105_3B
SAF

Figure 4. Autorotation RPM Correction Chart - 20%

September 11, 2004
Revised: October 18, 2013

**S-92A FLIGHT CHECK PROCEDURES CHECKLIST**

A/C NUMBER: _____

DATE: _____

PILOT: _____

COPILOT: _____

OBSERVERS: _____

AIRCRAFT BASIC WEIGHT: _____

TAKEOFF FUEL: _____

AIRCRAFT CENTER OF GRAVITY: _____

TAKEOFF WEATHER: OAT: _____

PA: _____

WIND: _____

MAINTENANCE PERFORMED: _____

PURPOSE OF FLIGHT: _____

FLT	DATE	GRD TIME	FLT TIME	CREW	REMARKS
1.					
2.					
3.					
TOTALS					

SIGNATURE OF PILOT: _____

DATE: _____



1.0 GROUND TEST PROCEDURES

1.1 EXTERNAL EMERGENCY LIGHTS	_____
1.2 FUEL SELECTORS	_____
1.3 STANDBY ATTITUDE INDICATOR	_____
1.4 BATTERY CHECKS	_____
1.5 SEAT BELT AND SHOULDER HARNESS	_____
1.6 SEAT ADJUSTMENT	_____
1.7 TAIL ROTOR PEDAL ADJUSTMENT	_____
1.8 THROTTLE QUADRANT	_____
1.9 PLACARDS	_____
1.10 EXTERNAL POWER	_____
1.11 APU START	_____
1.12 PARKING BRAKE	_____
1.13 ROTOR BRAKE	_____
1.14 HUMS	_____
1.15 CLOCK	_____
1.16 AIRCRAFT CONFIGURATION	_____
1.17 FLIGHT MANAGEMENT SYSTEM	_____
1.18 MULTIFUNCTION FLIGHT DISPLAYS	_____
1.19 SLEW CONTROLLER	_____
1.20 DISPLAY CONTROL PANEL	_____
1.21 REMOTE INSTRUMENT CONTROLLER	_____
1.22 REVERSION CONTROL PANEL	_____
1.23 MFD REVERSION	_____
1.24 CAUTION/ADVISORY CHECK	_____
1.25 AIRCRAFT GROSS WEIGHT	_____
1.26 FIRE DETECTION TEST	_____
1.27 FIRE EXTINGUISHER TEST	_____
1.28 ICS	_____
1.29 LIGHTS	_____
1.30 PITOT-STATIC AND OAT	_____
1.31 VHF RADIO	_____
1.32 GUARD RADIO	_____
1.33 ENHANCED GROUND PROXIMITY WARNING SYSTEM	_____
1.34 COCKPIT VOICE RECORDER/FLIGHT DATA RECORDER	_____
1.35 RADAR ALTIMETER	_____
1.36 HYDRAULIC LEAK ISOLATION TEST	_____
1.37 FLIGHT CONTROL LIMIT CHECK	_____
1.38 PRIMARY SERVOS CHECK	_____
1.39 BOOST SERVOS CHECK	_____
1.40 TRIM CHECKS	_____
1.41 AFCS PREFLIGHT BIT CHECK	_____
1.42 WINDSHIELD WIPER/WASHER	_____
1.43 PITOT-HEATER	_____



1.44 ENVIRONMENTAL CONTROL SYSTEM _____

2.0 RUN-UP TEST PROCEDURE

2.1	STARTER SYSTEM	_____
2.2	ENGINE START	_____
2.3	ENGINE CROSSBLEED START	_____
2.4	ENGINE CROSSBLEED MOTORING	_____
2.5	ROTOR ENGAGEMENT	_____
2.6	HYDRAULIC SYSTEM CHECK	_____
2.7	INITIAL RUN-UP CHECKS	_____
2.8	MGB OIL BYPASS CHECK	_____
2.9	DRIVE SYSTEM LEAK CHECK	_____
2.10	ENGINE ACCELERATION CHECK	_____
2.11	ENGINE BREAK-IN	_____
2.12	ELECTRICAL SYSTEM	_____
2.13	WINDSHIELD ANTI-ICE	_____
2.14	AFCS GROUND RUN CHECK	_____
2.15	AUTOMATIC VIBRATION CONTROL	_____
2.16	GROUND MAIN ROTOR TRACK AND BALANCE	_____
2.17	TAIL ROTOR BALANCE	_____
2.18	ENGINE HIGH SPEED SHAFT BALANCE	_____
2.19	ENGINE HIGH SPEED SHAFT DIAGNOSTIC	_____
2.20	ENGINE ANTI-ICE	_____
2.21	FUEL SELECTOR AND SHUTOFF VALVE	_____
2.22	DOOR	_____
2.23	RAMP	_____
2.24	ROTOR BRAKE BREAK-IN	_____
2.25	MAINTENANCE DATA COMPUTER	_____
2.26	GROUND ENGINE POWER ASSURANCE CHECK	_____
2.27	FADEC TRAINING MODE	_____
2.28	BMS	_____
2.29	HUMS	_____
2.30	WHEEL BRAKE BREAK-IN	_____

3.0 HOVER CHECKS

3.1	FLIGHT MAIN ROTOR TRACK	_____
3.2	WEIGHT ON WHEELS CHECK	_____
3.3	INITIAL HOVER CHECKS	_____
3.4	HOVER CONTROL POSITION CHECK	_____
3.5	COMPASS AND HSI CHECK	_____
3.6	AFCS HOVER CHECKS	_____



4.0 FLIGHT CHECKS

- 4.1 FLIGHT INSTRUMENTS
- 4.2 AUTOROTATION
 - COLLECTIVE SETTING 10%
 - COLLECTIVE SETTING 15%
 - COLLECTIVE SETTING 20%
 - HP
 - OAT
 - DENSITY ALTITUDE
 - FUEL
 - GROSS WEIGHT
 - ACTUAL NR
 - TARGET NR
- 4.3 LANDING GEAR CHECK
- 4.4 FLIGHT ENGINE POWER ASSURANCE CHECK
- 4.5 AFCS FLIGHT CHECKS
- 4.6 ENHANCED GROUND PROXIMITY WARNING SYSTEM
- 4.7 VH CHECK

5.0 EQUIPMENT CHECKS

- 5.1 TCAS
- 5.2 FLOTATION
- 5.3 LIFE RAFT
- 5.4 RESCUE HOIST
- 5.5 CARGO HOOK
- 5.6 RIPS
- 5.7 WEATHER RADAR
- 5.8 LIGHTNING STRIKE SENSOR
- 5.9 DIGMAP
- 5.10 SAR AFCS
- 5.11 RIG APPROACH

